From <u>**method**</u> to <u>**usable system**</u>
Human-Centered NLP

Billy Xuanming Zhang

Slides credits: Philippe Laban and Diyi Yang

NLP in the Age of LLMs

Recent Progress:

- Machine translation
- Sentiment analysis
- Conversational agent
- Question answering

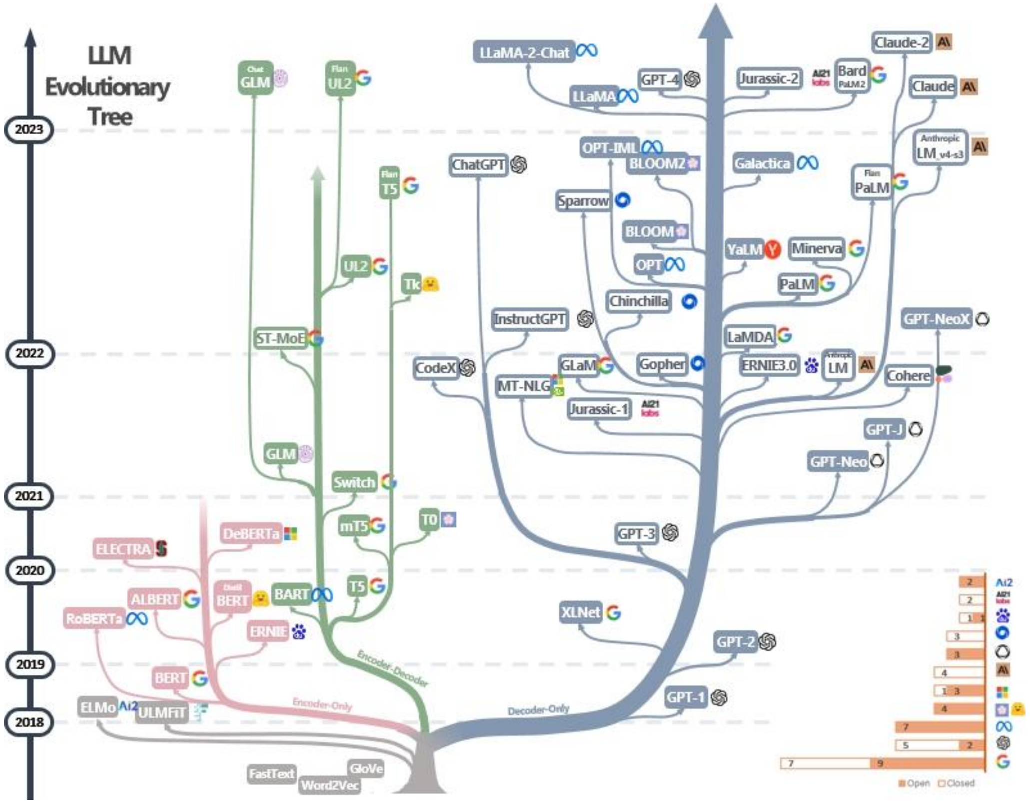


Image credit to https://github.com/Mooler0410/LLMsPracticalGuide

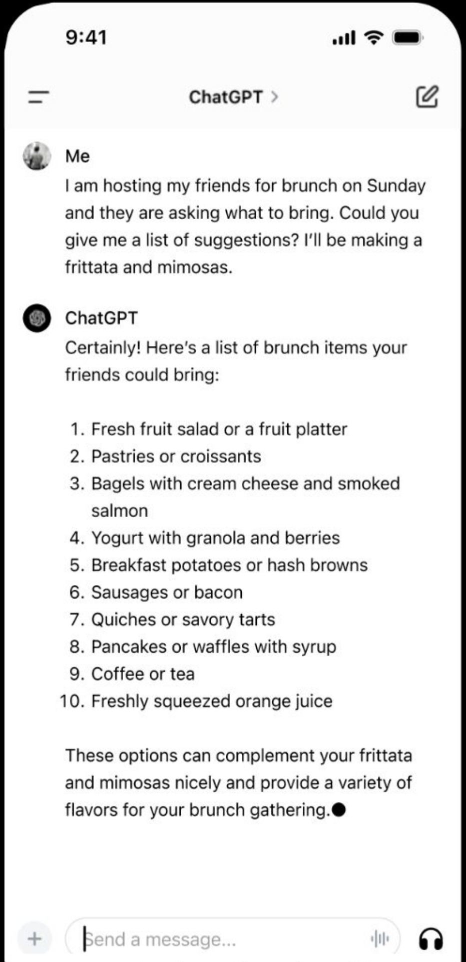


Image credit to https://openai.com/chatgpt

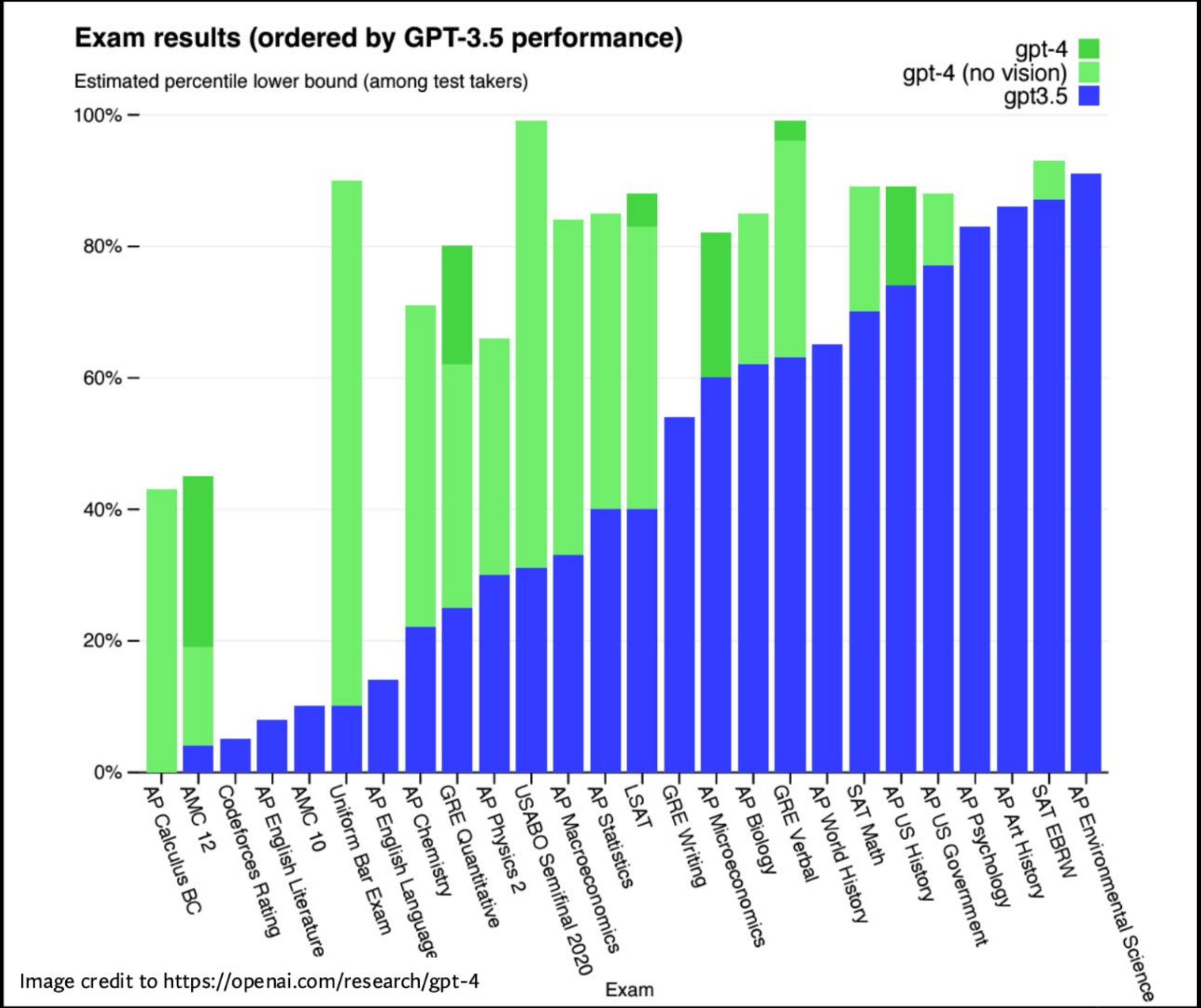
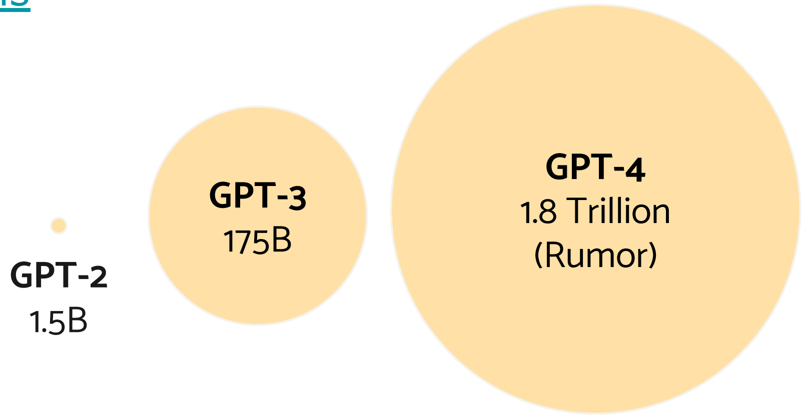


Image credit to https://openai.com/research/gpt-4

In the era of LLMs …

- LLMs have significantly advanced various NLP tasks, surpassing traditional models and narrowing the gap between machine and human capabilities
  - General NLP Benchmarks:
    - GLUE, SuperGLUE, MT-Bench
  - Question Answering:
    - SQuAD, TriviaQA, Natural Questions
  - Math and Reasoning
    - Big-Bench, GSM8K
  - …



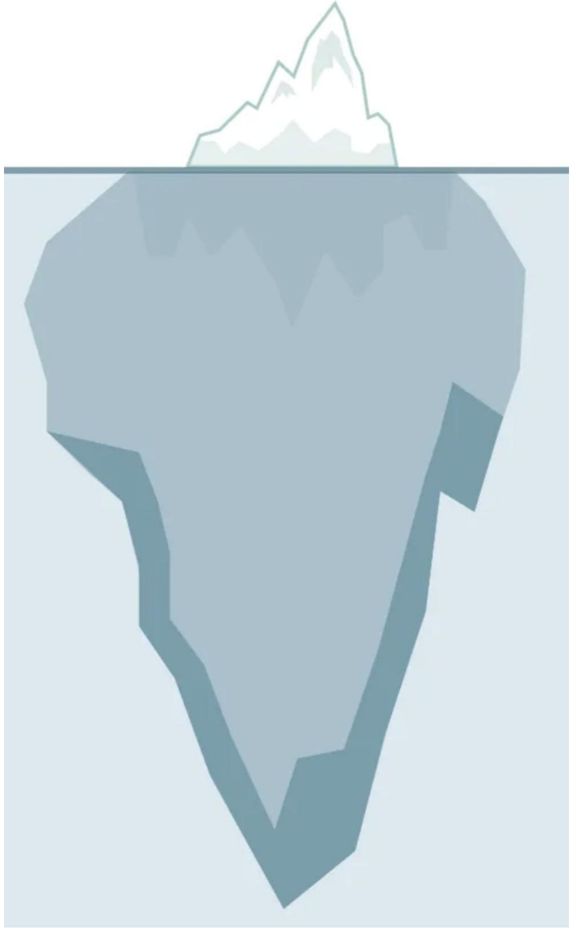
It seems like LLMs can already perform well on various NLP tasks …

Can LLMs solve all NLP tasks?

What keeps us from using it in practical settings?

(you might be too shy to ask…)

Going from a <u>**method**</u> to a <u>**usable system**</u>: Human-Centered NLP



Overview

Part 1: An introduction to Human-Centered NLP

Part 2: Case studies with two human-centered systems

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Part 1: An introduction to Human-Centered NLP

Part 2: Case studies with two human-centered systems

What is Human-Centered NLP?

"Human-centered NLP involves designing and developing NLP systems in a way that is attuned to the needs and preferences of human users, and that considers the ethical and social implications of these systems."

— Diyi Yang (Stanford)

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What is Human-Centered NLP?

It touches multiple NLP dev stages.

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What is Human-Centered NLP?

It concerns NLP system, which goes beyond just the model - also includes e.g. user interfaces on top of the model.

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It needs to be optimized for humans.

"Optimize for humans" require careful ethical concerns.

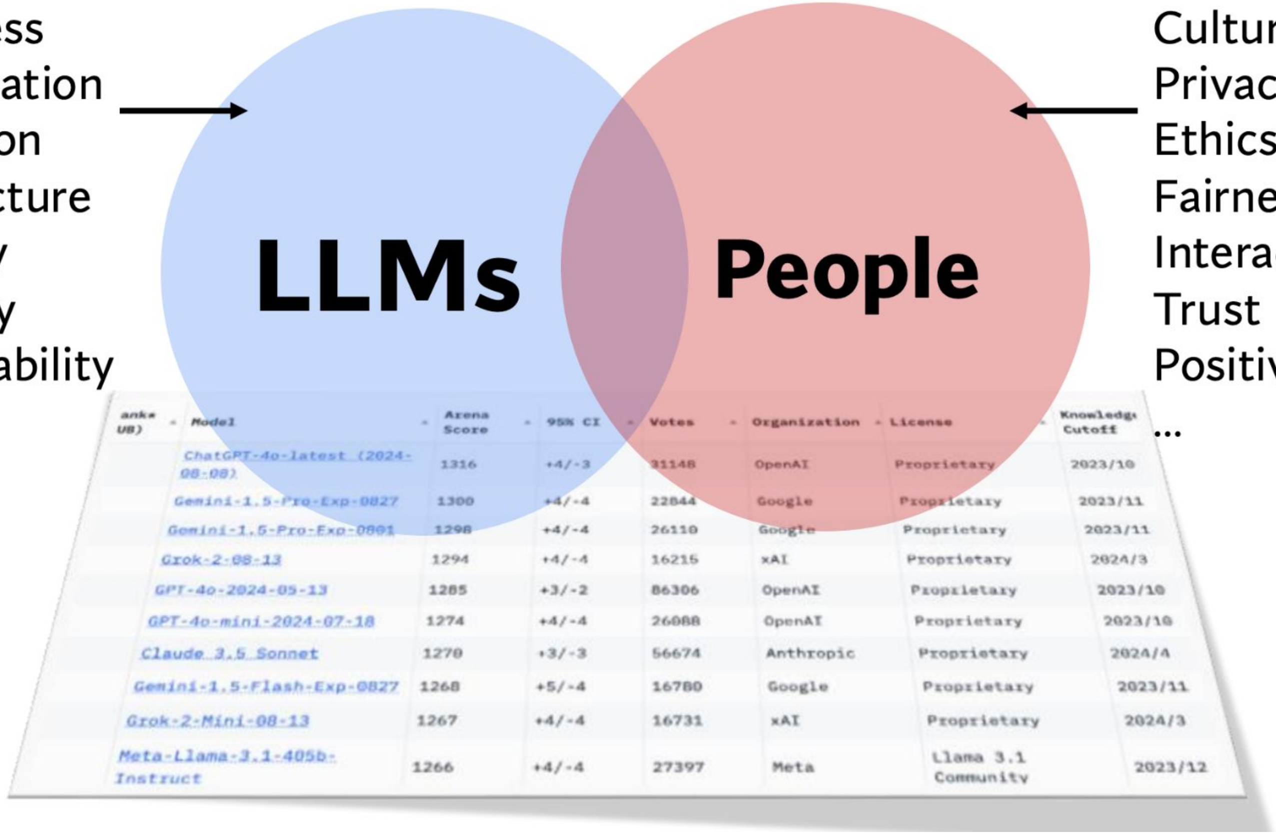
Human-Centered NLP - Beyond the ML/NLP Method

Situation: I have built a cool NLP model and I want people to use it!

What should I consider when building the system that uses my NLP method?

Reasoning
Benchmarking
Robustness
Generalization
Verification
Infrastructure
Efficiency
Scalability
Interpretability
...

Personality
Social Factors
Culture and Value
Privacy
Ethics
Fairness
Interaction
Trust
Positive Impact
...



ank* UB)	Model	Arena Score	95% CI	Votes	Organization	License	Knowledge Cutoff
	ChatGPT-4o-latest (2024-08-08)	1316	+4/-3	31148	OpenAI	Proprietary	2023/10
	Gemini-1.5-Pro-Exp-0827	1300	+4/-4	22844	Google	Proprietary	2023/11
	Gemini-1.5-Pro-Exp-0801	1298	+4/-4	26110	Google	Proprietary	2023/11
	Grok-2-08-13	1294	+4/-4	16215	xAI	Proprietary	2024/3
	GPT-4o-2024-05-13	1285	+3/-2	86306	OpenAI	Proprietary	2023/10
	GPT-4o-mini-2024-07-18	1274	+4/-4	26088	OpenAI	Proprietary	2023/10
	Claude 3.5 Sonnet	1270	+3/-3	56674	Anthropic	Proprietary	2024/4
	Gemini-1.5-Flash-Exp-0827	1268	+5/-4	16780	Google	Proprietary	2023/11
	Grok-2-Mini-08-13	1267	+4/-4	16731	xAI	Proprietary	2024/3
	Meta-Llama-3.1-405b-Instruct	1266	+4/-4	27397	Meta	Llama 3.1 Community	2023/12

An Intro to Human-Centered NLP

PART 1

Beyond the ML Method

Situation: I have built a cool NLP model and I want to people to use it!

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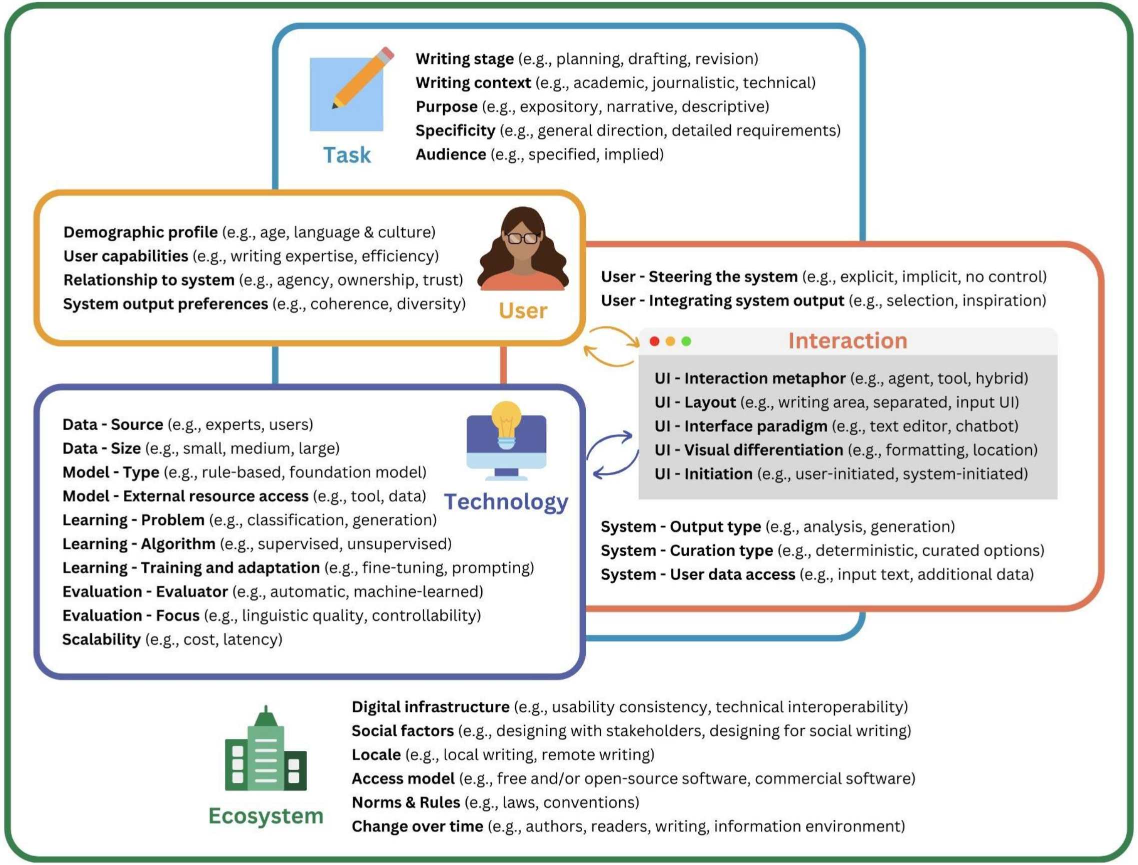
PART 2

Usability Testing Crash Course

Situation: I have built a system that uses my NLP model. How do if “it works” for users?

A deep-dive into two usability studies of an LLM-based text-editing system.



Technology is only one aspect to consider when building a System.

Important to also consider what the **Task** is, who the **User** is, what the **Interaction** is, and how the **Ecosystem** works.

See this great paper surveying 115 papers in the space!

A Design Space for Intelligent and Interactive Writing Assistants, Lee et al. – CHI 2024

Task: What is the user trying to accomplish?

Example: consider the Writing Stage of the user.

What stage of writing does the system support?

(1) Idea Generation, (2) Planning, (3) Drafting, or (4) Revision?

Task: What is the user trying to accomplish?

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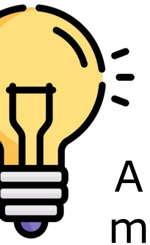


A chat interface might be most adequate for these...

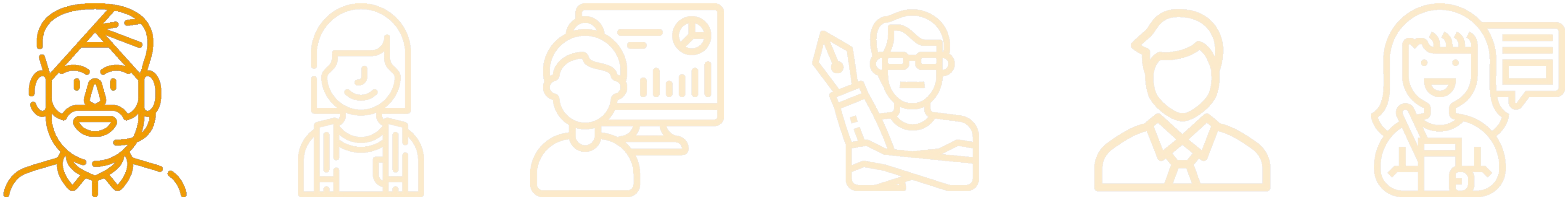
... auto-complete for this

... & a Grammarly-style text editor for this

(see the paper for other task considerations: Writing Context, Purpose, Specificity, and Audience)

User: Who is the target user of the system?

Example: What are the target User Capabilities?



An international Wikipedia Editor might need help with American English rules.

(see the paper for other user considerations: Demographics, Relationship to System, System Output Preferences)

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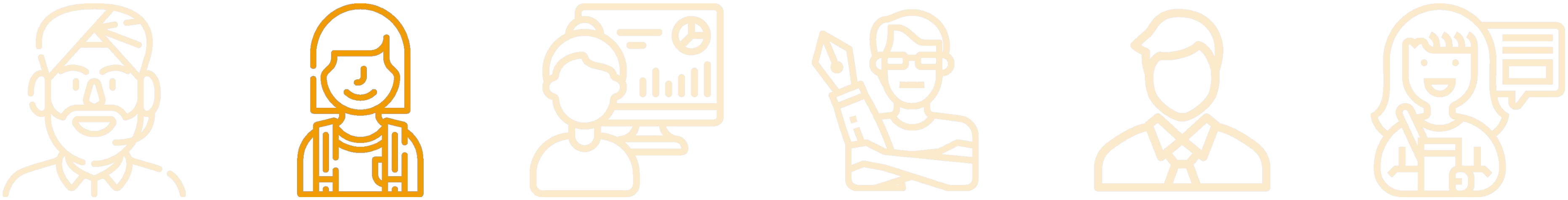

A 6th Grade Student might need help spotting grammar & spelling issues.

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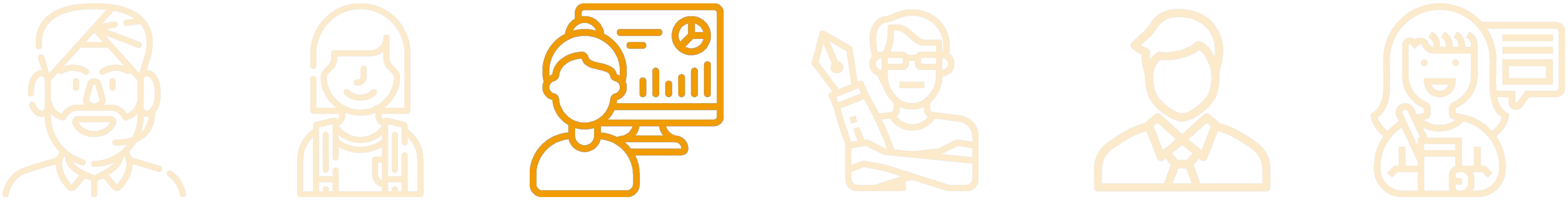


A Ph.D. Student might need help making sure technical terminology is accurate.

(see the paper for other user considerations: Demographics, Relationship to System, System Output Preferences)

User: Who is the target user of the system?

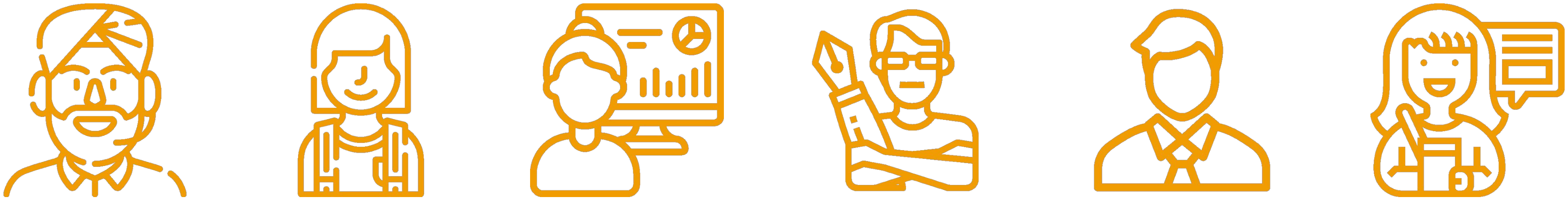
Example: What are the target <u>User Capabilities</u>?

A professional writer might not want the system to make changes affecting their style

(see the paper for other user considerations: Demographics, Relationship to System, System Output Preferences)

User: Who is the target user of the system?

Example: What are the target <u>User Capabilities</u>?



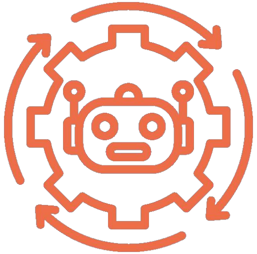
The user's capabilities and needs should be considered during system design!

(see the paper for other user considerations: Demographics, Relationship to System, System Output Preferences)

Interaction: How do User, User Interface, and System interact?

Example: How is the system output triggered? (Initiation)

User-Initiated
Reactive



System-Initiated
Proactive

(see the paper for other interaction considerations: Interface Paradigm, Visual Differentiation, Steering, Integration)

Interaction: How do User, User Interface, and System interact?

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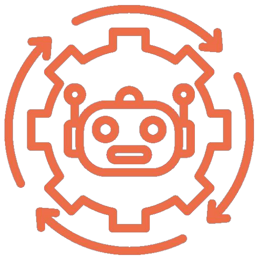
User-Initiated
Reactive

More natural for
Idea Generation & Planning
(ask for what you want)



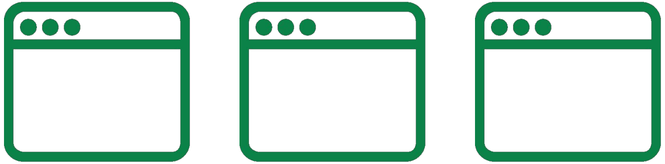
System-Initiated
Proactive

More natural for Revision, Grammar, Typos (continuous scan & fix)

(see the paper for other interaction considerations: Interface Paradigm, Visual Differentiation, Steering, Integration)

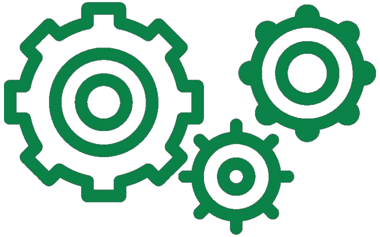
Ecosystem: Does the system fit in the overall ecosystem?

Example: What compatibility issues does the system consider?



Usability
Consistency

Does the system align with other systems in terms of usability?



Technical
Operability

With external services (APIs, tangible world, etc.)

(see the paper for other interaction considerations: Access Model, Social Factors, Locale, Norms, Change over Time)

Task, User, Interaction, Ecosystem

...

What if I just want to build cool NLP models?

Good news: You do not need to "check all the boxes".

But building useful systems likely requires considering some of these.

Task, User, Interaction, Ecosystem

...

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Good news: You do not need to "check all the boxes".

But building useful systems likely requires considering some of these.

We will see it in action later!

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Task, User, Interaction, Ecosystem, …

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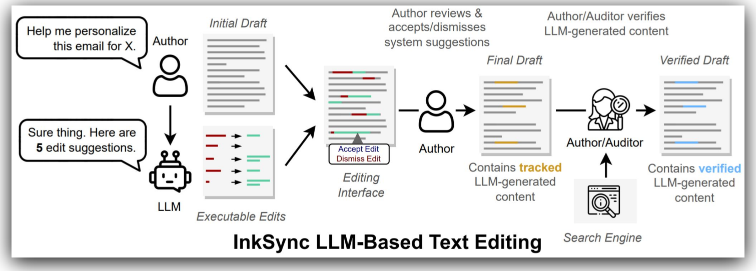
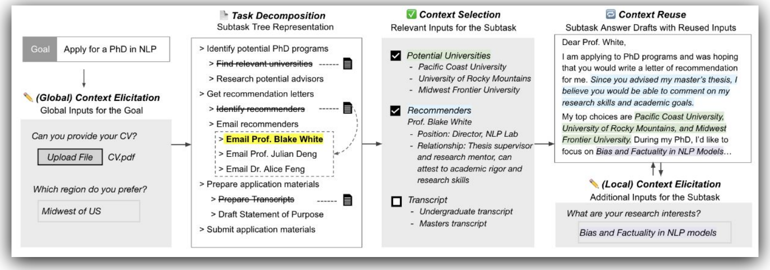


[InkSync](Laban et al., 2024)



[JumpStarter](Zhang and Wang et al., 2025)

InkSync: Executable and Verifiable Text-Editing with LLMs

Philippe Laban, Jesse Vig, Marti Hearst, Caiming Xiong, Chien-Sheng Wu

Salesforce Research

UIST 2024

Let’s see it in action:

InkSync

Executable and Verifiable Text-Editing with LLMs

Philippe Laban, Jesse Vig, Marti Hearst, Caiming Xiong, Chien-Sheng Wu

UIST 2024

Motivation: Imagine that you are drafting an email

In the past… you have probably sent an email draft to ChatGPT, asking for feedback on:

- How to change to the (+ formal)
- Fix typos/grammar
- Insert additional details
- Etc.

You

Help me make my email sound more formal and professional. Also add details about things to do in Rome.

` ` `

Subject: Plan your next vacation.

Hey Bob,

How have you been? Are you tired and wanna go on a vacation? I have special invitation to Rome, here are the details.

Experience vibrant cultur and explore the stunning landscapes the place has to offer.

Dont miss out on this fantastc opportunity! Reach out to us now to book your escape. Your dream vacation awaits!

Cheers,
InkSync Agency

` ` `

Motivation: LLM Typical Reply to Rewriting

The LLM typically replies with a new version of the document.

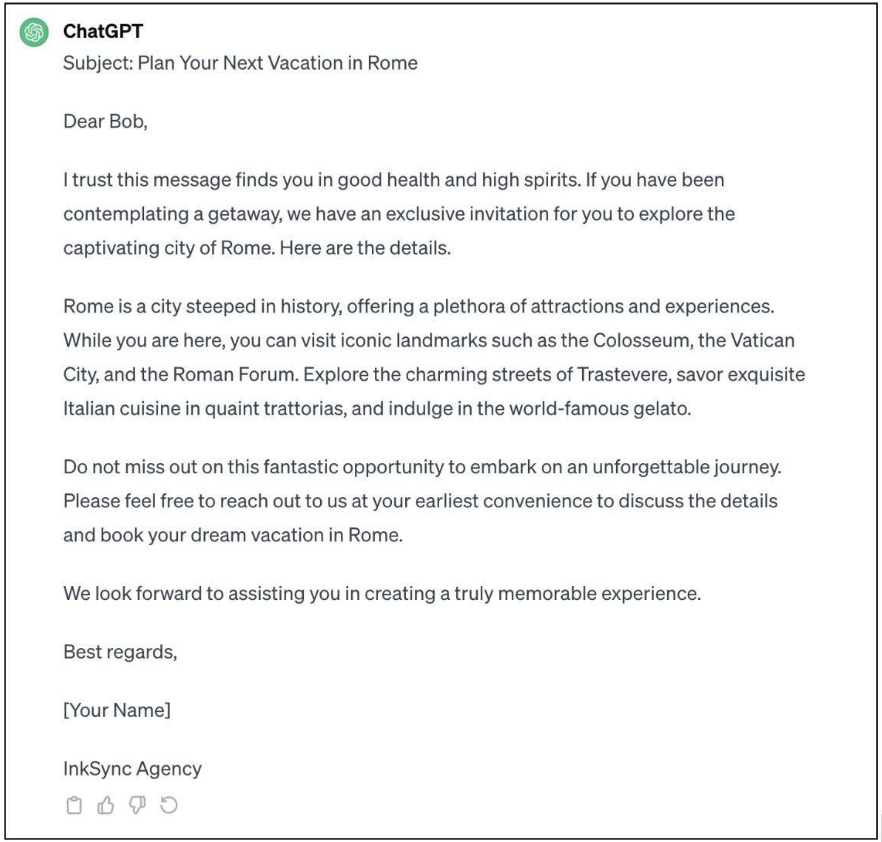
Pros:

- Ready-to-use
- Convenient to copy-paste

Cons:

- Editing suggestions are implicit
- Not easy to accept subset of edit suggestions

We surveyed knowledge workers: most who use LLMs for rewriting frequently accept LLM rewrites verbatim. Most look at suggestions, and decide which to include in their editor.



ChatGPT

Subject: Plan Your Next Vacation in Rome

Dear Bob,

I trust this message finds you in good health and high spirits. If you have been contemplating a getaway, we have an exclusive invitation for you to explore the captivating city of Rome. Here are the details.

Rome is a city steeped in history, offering a plethora of attractions and experiences. While you are here, you can visit iconic landmarks such as the Colosseum, the Vatican City, and the Roman Forum. Explore the charming streets of Trastevere, savor exquisite Italian cuisine in quaint trattorias, and indulge in the world-famous gelato.

Do not miss out on this fantastic opportunity to embark on an unforgettable journey. Please feel free to reach out to us at your earliest convenience to discuss the details and book your dream vacation in Rome.

We look forward to assisting you in creating a truly memorable experience.

Best regards,

[Your Name]

InkSync Agency

Motivation: Reverse-Engineering the LLM Output

Using a string-alignment algorithm (Levenshtein), we can recover the exact edits proposed by the LLM: the characters inserted and ~~deleted~~.

This makes the editing suggestions more **explicit**.

But such string alignment visualization is not common in chat interfaces, and taking editing actions based on this would require manual work, that could be error-prone.

Subject: Plan ~~y~~Your ~~n~~Next ~~v~~Vacation in Rome.

~~Hey~~Dear Bob,

~~How have you been?~~ I trust this message finds you in good health and high spirits. ~~Are you tired and wanna go on a vacation?~~ If you have been contemplating a getaway, ~~I have special~~ we have an exclusive invitation to explore the captivating city of Rome,~~.~~ ~~h~~Here are the details.

~~Experience vibrant cultur and explore the stunning landscapes the place has to offer.~~ Rome is a city steeped in history, offering a plethora of attractions and experiences. While you are here, you can visit iconic landmarks such as the Colosseum, the Vatican City, and the Roman Forum. Explore the charming streets of Trastevere, savor exquisite Italian cuisine in quaint trattorias, and indulge in the world-famous gelato.

Do~~nt~~ not miss out on this fantastic opportunity to embark on an unforgettable journey~~!~~. Please feel free to ~~R~~reach out to us ~~now~~at your earliest convenience to discuss the details and book your ~~escape~~ dream vacation in Rome.

~~Your dream vacation awaits!~~ We look forward to assisting you in creating a truly memorable experience.

~~Cheers~~Kind Regards,
[Your Name]
InkSync Agency

Motivation: What if we ask for explicit edits directly?

We could ask the LLM to provide its feedback in bullet-point form? Adding this to the prompt:

> Do not rewrite the document. Instead, list out exactly all the editing changes you would make to the document.

Sorry for the small font, answer is **much longer.**

Pros:

- LLM provides explanations/reasons.
- Transparent: know how many edits are suggested.

Con: Still hard to integrate, requires manual work, so nobody does this.

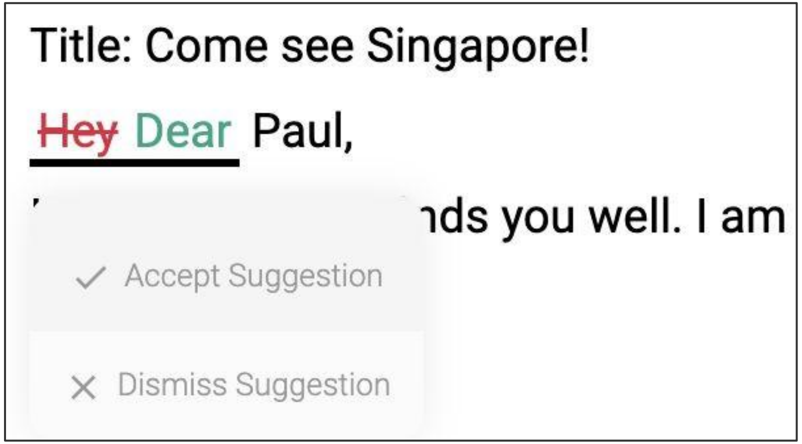
> **ChatGPT**
>
> Here are the editing changes I would make to your email to make it sound more formal and professional:
>
> 1. Change the subject line to be more descriptive and professional:
>   Subject: Discover Rome: Your Next Vacation Destination
> 2. Address the recipient more formally:
>   Change "Hey Bob," to "Dear Bob,"
> 3. Improve the opening sentence to sound more professional:
>   Change "How have you been?" to "I hope this message finds you well."
> 4. Make the invitation more formal:
>   Change "I have special invitation to Rome, here are the details." to "We are pleased to extend an exclusive invitation for you to visit Rome. Please find the details below."
> 5. Enhance the description of Rome:
>   Change "Experience vibrant cultur and explore the stunning landscapes the place has to offer." to "Immerse yourself in the vibrant culture and explore the breathtaking landscapes that Rome has to offer."
> 6. Correct the spelling and grammar errors:
>   - Change "cultu**r**" to "culture."
>   - Change "fantastc" to "fantastic."
> 7. Include more information about Rome's attractions:
>   Add a brief paragraph about popular things to do in Rome, such as visiting the Colosseum, exploring the Vatican City, trying authentic Italian cuisine, and strolling through historic neighborhoods like Trastevere.
> 8. Encourage action in a professional manner:
>   Change "Dont miss out on this fantastc opportunity! Reach out to us now to book your escape." to "Don't miss out on this fantastic opportunity! We encourage you to contact us at your earliest convenience to secure your dream vacation."
> 9. Sign off more formally:
>   Change "Cheers, InkSync Agency" to "Best regards, InkSync Agency"
>
> By making these changes, your email will appear more formal and professional while also providing additional information about things to do in Rome to entice the recipient.

In InkSync, we go a step further, and ask the LLM to generate executable edits, in a standardized edit language.

Executable Edit Language - Example Edit

```
{
  "original_text":    "Hey Paul",
  "replace_text":     "Dear Paul",
  "replace_all":"0",
  "component":        "chat",
  "new_info":   "0"
}
```

Executable edits are first-class objects in the text editor, which the user can review, accept & dismiss in one click.



InkSync Demo!

InkSync has **four components** that can produce executable edits:

CHAT

COMMENT

MARKERS

BRAINSTORM

InkSync Demo!

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CHAT

COMMENT

MARKERS

BRAINSTORM

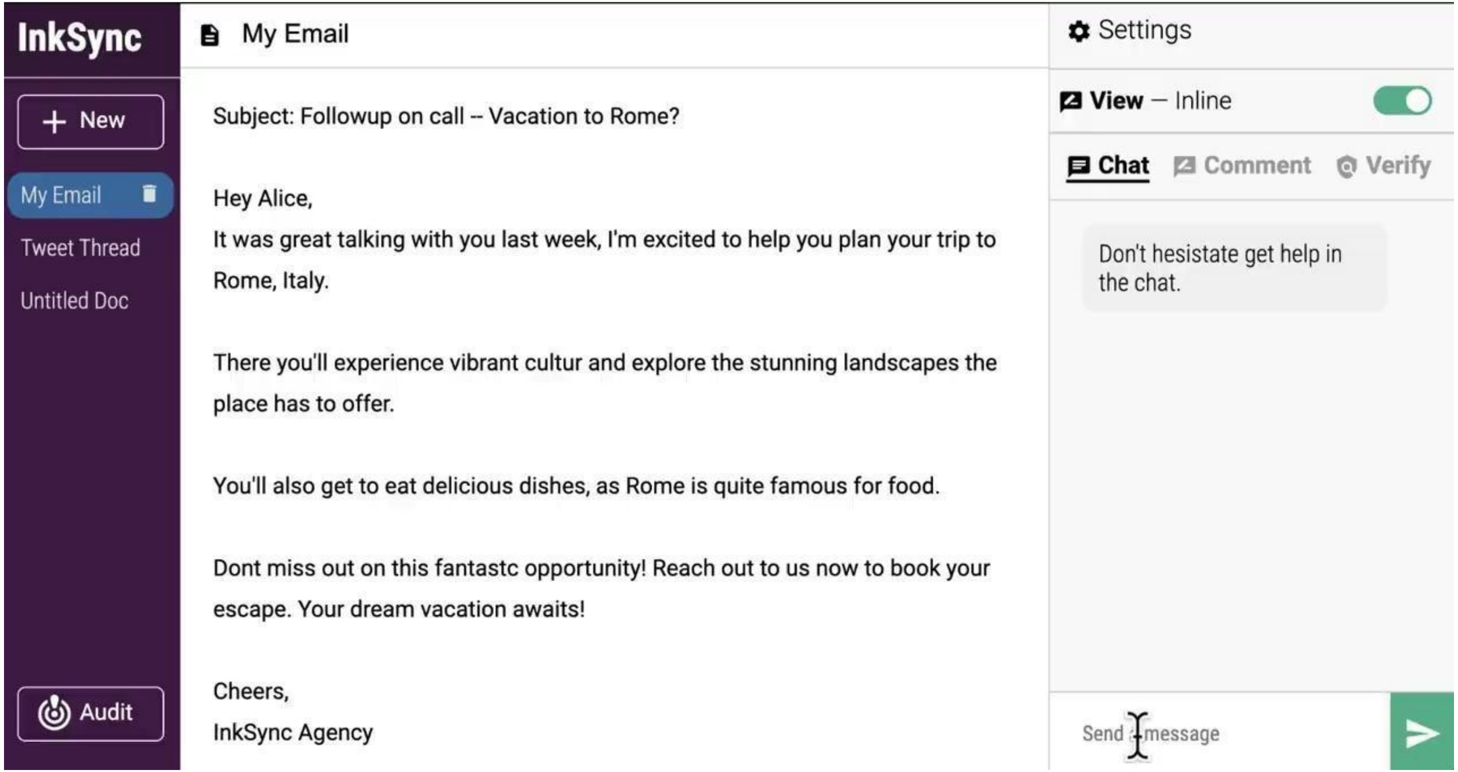
In this talk: we'll look at two of them, see paper/demo for other two.

InkSync Demo

CHA
T

1. The user asks for editing help
2. The LLM replies:
   2a. In plain text (in the chat)
   2b. With a list of executable edits (viewed in the editor)

Because Chat is LLM-based, it can handle a broad range of editing tasks.



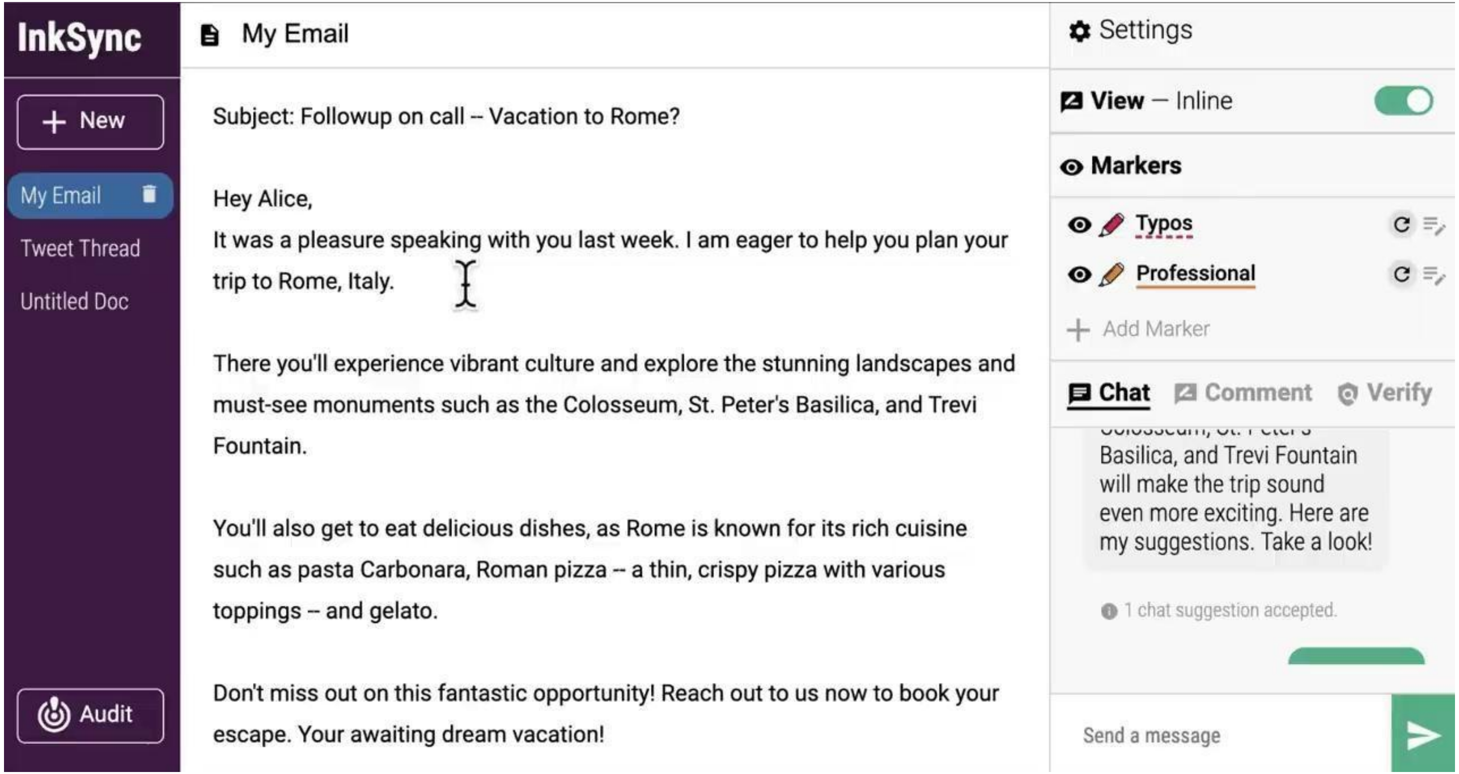
InkSync Demo

MARKERS

1. The user creates a Marker once. (e.g.: *look out for Typos*)

2. During editing, Markers continuously suggests executable edits, running in the background.

Markers = proactive
Chat = reactive



How does InkSync fit in the Design Space?

Aspect	Dimensions	Tags

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Task	Writing Stage	Drafting and Revision ~~(not Idea Generation G Planning)~~

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User	Relationship to System	Agency (easy to accept/dismiss individual suggestions), Trust (Factuality Checking)

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Interaction	UI — Interface Paradigm	Text Editor with Chat on the Side
Interaction	UI — Visual Differentiation	Formatting (Red/Green Styling)
Interaction	UI — Initiation	Both Reactive (Chat) and Proactive (Markers)

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Interaction	UI — Initiation	Both Reactive (Chat) and Proactive (Markers)
Ecosystem	Model Access	Does not allow for model change
Ecosystem	Usability Consistency	Visual Resemblance to other Text Suggestion systems

<u>In summary:</u> All dimensions might not be relevant to every project. But considering where a project fits within the space can help situate the work & come up with ideas.

An Intro to Human-Centered NLP

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Human-Centered NLP - Usability Testing

Situation: I have built a system that uses my NLP model. How to test if “it works” for users? How do we know if users like it?

The core of Human-Centered systems:
User Studies

Human-Centered NLP - Usability Testing

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The core of Human-Centered systems:
User Studies

5 pieces of advice on designing user studies

Advice #1: Consider the Task & Participants Carefully

✔ Select participants that would find the task "relevant".

Recruit on MTurk for a task that requires specialized skills

✔ Think carefully of total task duration:

- Enough time for meaningful interaction & feedback
- Not too long to avoid burnout / skipping

✔ Plan for time to train/onboard participants on how to use the interface

InkSync Study 1: Task Setup

Advice #1: Consider the Task & Participants Carefully

Task: Participants are given an email template from the "*InkSync Travel Agency*"
They have 6 minutes to:

- Objective A: Make sure it sounds professional / doesn't have typos
- Objective B: Customize it to a travel destination / customer persona

They will use different kinds of interfaces with and without assistance.

Advice #2: Select Broad Study Conditions

- ✔ Make sure to select **baselines** and “**oracle**” settings when possible.

Don’t just compare variations of your system

- ✔ Map potential differences in conditions to research questions.

Don’t overdo it. More conditions = more participants

- ✔ Decide on study design:
  - Between-subjects: Each participant interacts with **one** condition
  - Within-subjects: Each participant interacts with **all** conditions

InkSync Study 1: Study Conditions

Advice #2: Select Broad Study Conditions

Six conditions:

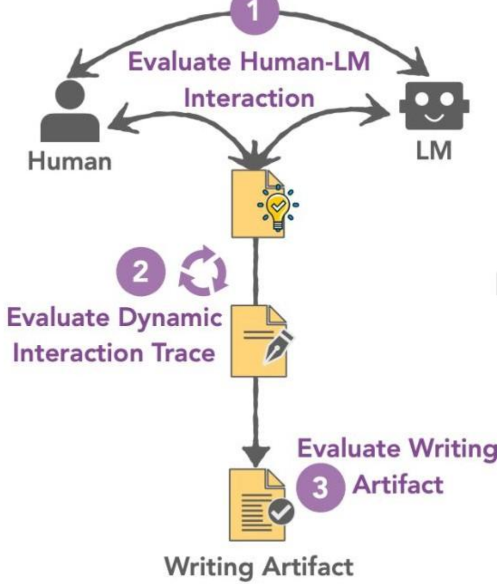
Manual Editing (Baseline)	Non-Executable Chat (Baseline)	Chat Only
Markers Only	Local Only	InkSync 4-Components

- Recruit ~100 participants.
- Each completing 3 sessions (in random order) -> total 300 editing sessions
- Intro Material (2-min) + 3 x 6-min editing + 2-min survey + 5-min buffer
  (you always need to include buffer time, unexpected things happen)

Advice #3: Include Redundancy in Evaluation Signal

- ✔ Evaluate Interaction Logs
- ✔ Evaluate "Product/Artifact" of Study
- ✔ Ask Participants for Preference/Opinion

Don't: Treat *Participants* as *Annotators.*
You will throw away a lot of good signal.



(a) Parachute Overview

See great paper by Hua Shen et. al, 2023
[Parachute: Evaluating Interactive Human-LM Co-writing Systems]

InkSync Study 1: Evaluation Setup

Advice #3: Include Redundancy in Evaluation Signal

INTERACTION LOGS	ARTIFACT ANALYSIS	PARTICIPANT OPINION
Editing Efficiency (Levenshtein)	Objectives A/B Scoring	Likert Usability Questions
Component Usage Popularity	Hallucination Analysis	Condition Preference Ranking
	Divergence Analysis	

Redundancy = confirmation

InkSync Study 1: Example Completed Task

Initial Email Template - Study 1 (1 of 3)

Subject: Plan your next vacation.

Hey [Customer's Name],

How have you been? Are you tired and wanna go on a vacation? I have special invitation to [City Destination], here are the details.

Experience vibrant cultur and explore the stunning landscapes the place has to offer.

Dont miss out on this fantastc opportunity! Reach out to us now to book your escape. Your dream vacation awaits!

Cheers,
InkSync Agency

Gloria is traveling with 2 children

Destination: Egypt

Subject: ~~Your~~Plan The Next ~~Unforgettable Getaway!~~ Trip You Won't Forget to Egypt!

~~Hey~~Dear ~~[Customer's Name]~~Gloria,

~~How's life treating you?~~ Are you ~~F~~feeling exhausted and in need of an escape~~s~~? Well, ~~do~~ we have ~~a surprise~~ an exciting opportunity for you! We're extending a special invitation just for you to experience the trip of a lifetime to Egypt. You can visit the stunning Pyramids of Giza, one of the original seven wonders of ~~[City Destination].~~ the ancient world! ~~Prepare to~~ You will surely be amazed!

~~Get ready~~ Prepare to immerse yourself ~~of~~ in a world of rich culture and behold the breathtak~~er~~ing landscapes that await~~s~~ you. It's a place where ~~dreams come true!~~ memories are created and is an ideal destination for families. There are engaging experiences tailored for all ages, such as camel- riding around the Giza Plateau or a family cruise down the Nile River.

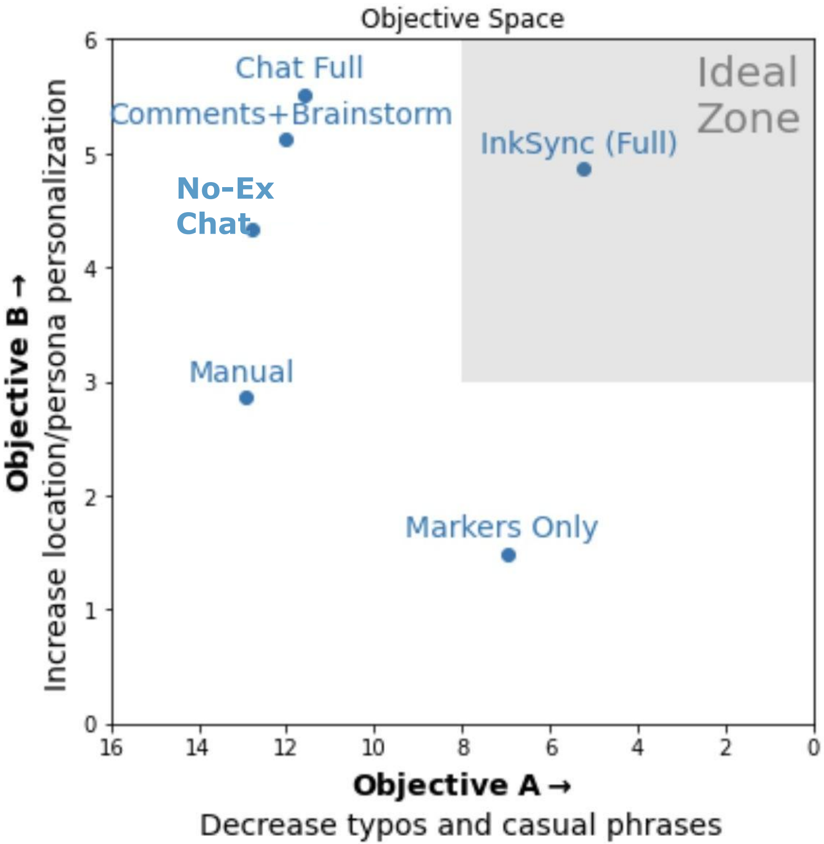
Imagine yourself wandering through the streets, immersing yourself in local tradition, ~~and discovering hidden gems that will make your back home friends envious.~~ visiting local markets, experiencing new culinary delights, snorkeling in the Red Sea, and even hot air ballooning in Luxor.

~~Don't let~~ Seize this incredible opportunity ~~slip through your fingers~~ without delay. Reach out to us now to secure your spot for an extraordinary vacation in Egypt that you'll cherish forever. Your perfect getaway, complete with a taste of Egypt's delicious cuisine with its delicious dishes like Koshari, Molokhia, and Ta'meya, is just a call away!

~~Cheers~~Kind Regards,
InkSync Agency

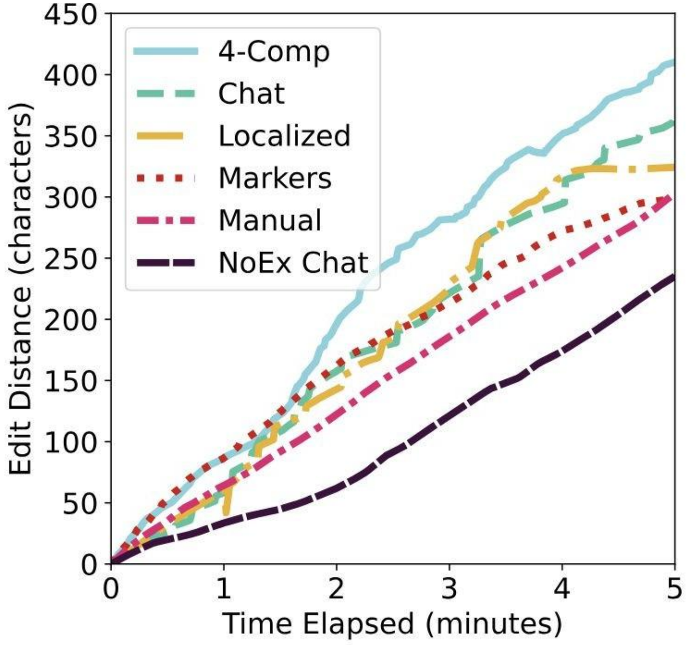
Legend: Manual Editing Markers Chat Comment Brainstorm

InkSync Study 1: Objectives A/B Scoring



- "Markers"-like interactions help with Objective A: constantly identifying & remove typos / casual phrases.
- "Chat"-like interactions (even non-executable) help with Objective B: customization / idea generation.
- Combining both (InkSync Full) leads to the best of both: components are complementary.

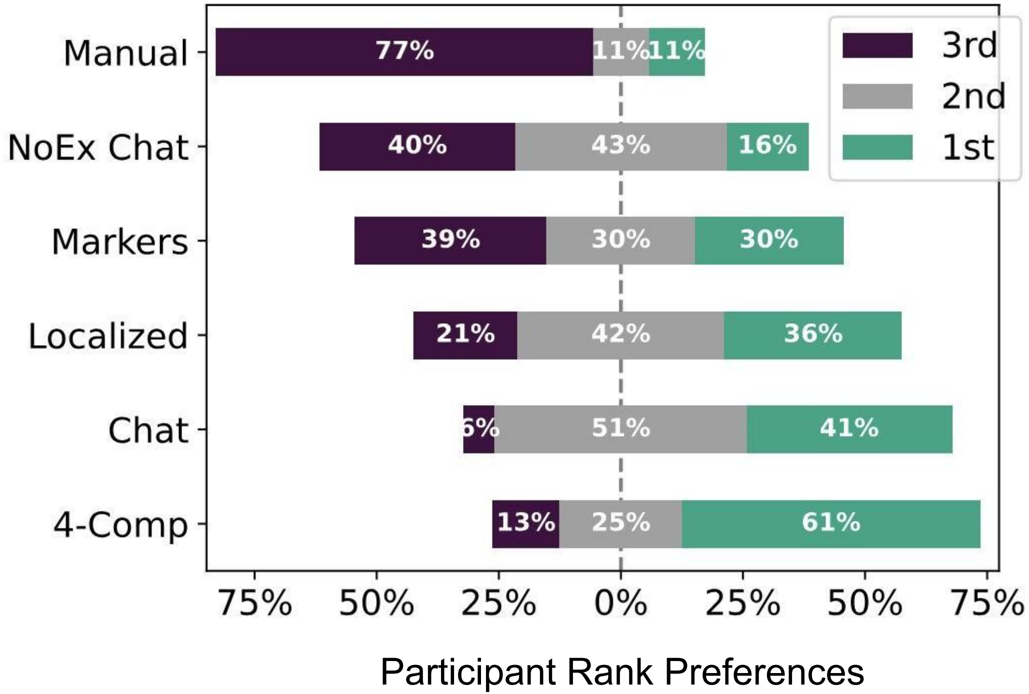
InkSync Study 1: Editing Efficiency



Edit Efficiency Analysis

- Participants with executable edits lead to faster/more efficient editing.
- NoEx Chat even slower than Manual (!) -> divided attention, more effort required.
- 4-Comp (Full interface) leads to fastest editing -> complementary components.

InkSync Study 1: Condition Preference Ranking



Participants ranked the 3 interfaces they used.

4-Comp most preferred, followed by Chat. NoEx Chat G Manual least preferred.

InkSync Study 2: Hallucination Analysis

Manual analysis of ~300 editing sessions reveals:

- 28% of edited emails contain a minor factual error
    - *e.g., saying that Paris is an affordable European city*
- 15% of edited emails contain a major factual error
    - *e.g., recommending a visit to a Zoo that closed in 2017*

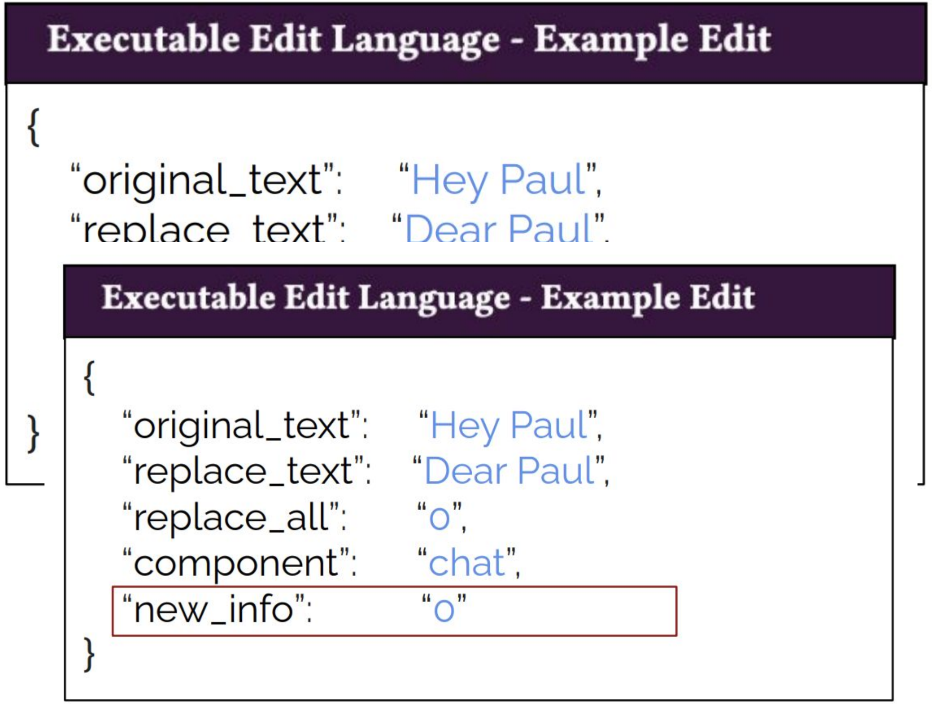
To mitigate such risks, we implement a framework that builds on top of executable edits:

WARN - VERIFY - AUDIT

When the LLM generates an executable edit, it must populate a *new_info* key: does the edit introdu new information or not?

When new information is introduced, there's a risk of hallucination and the user is <u>warned.</u>

(a manual analysis confirms GPT4 is ~97.5% accurate at this task)

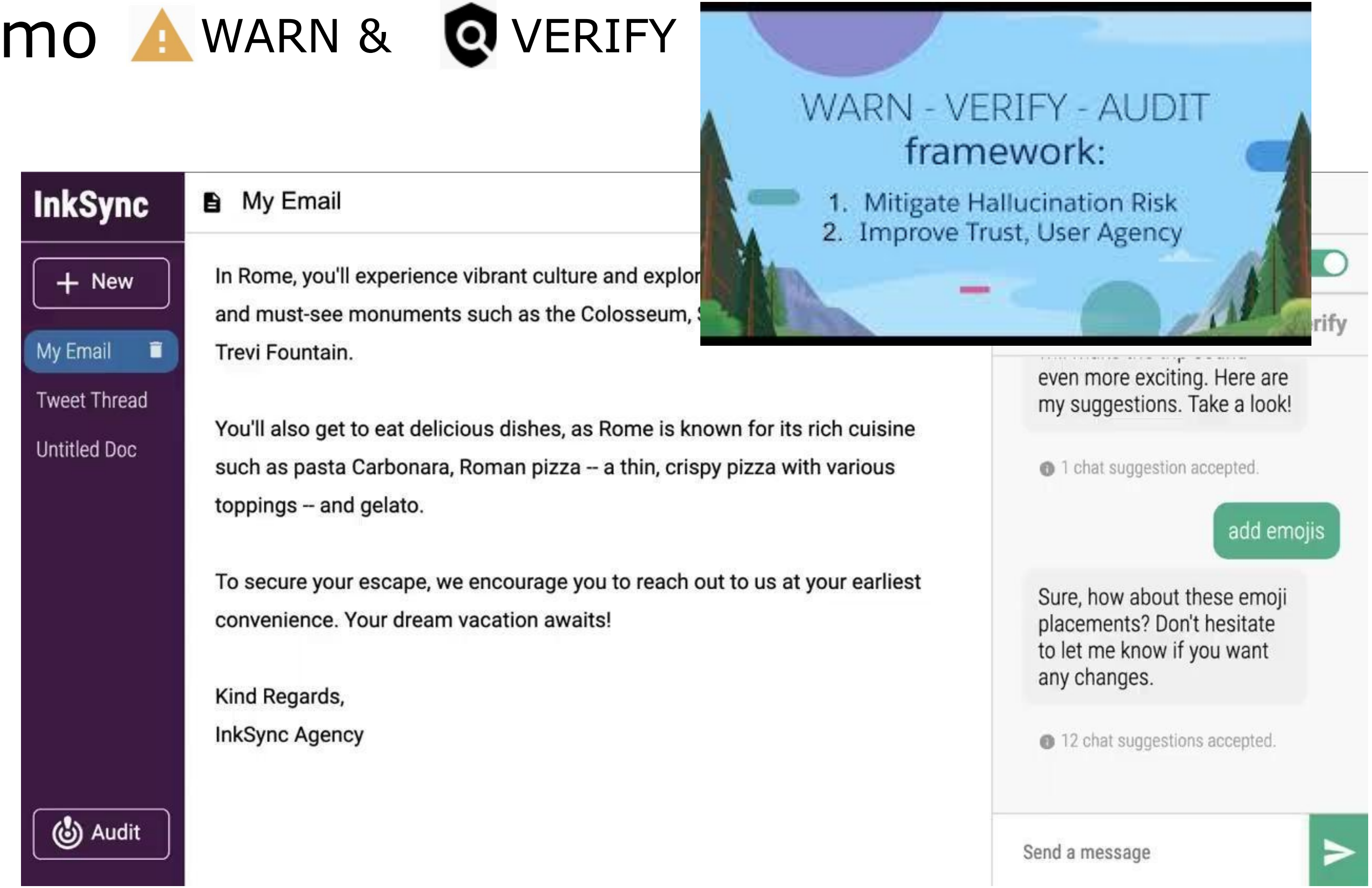


InkSync Demo WARN & VERIFY

When a warning is shown, the user is given an option to **<u>verify</u>** the edit.

To verify the edit, the LLM generates **search engine queries**, that the user can visit to assess the edit's veracity.

Verification is human-in-the-loop by design.

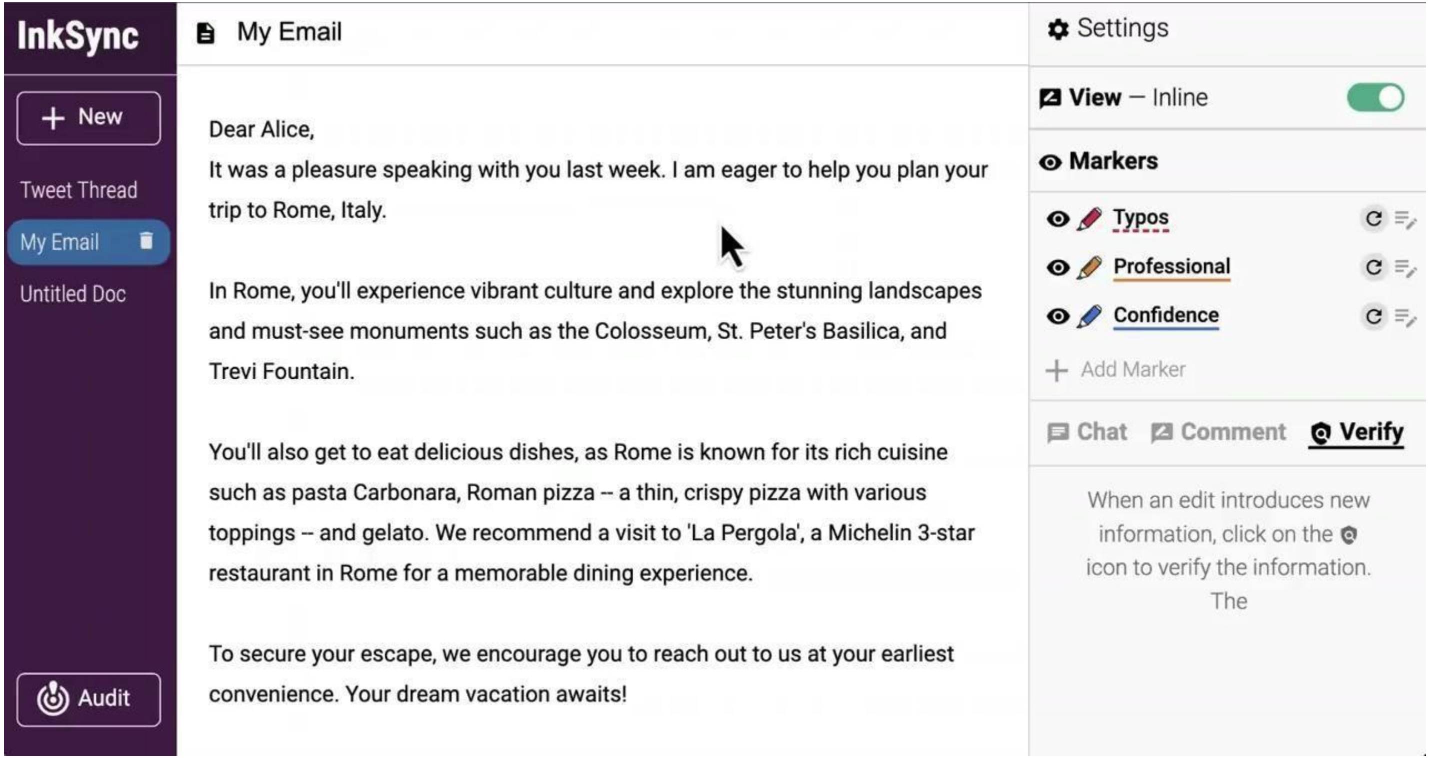


InkSync Demo AUDIT

Because edits are **executed**, we can trace each character's origin (LLM-gen vs. human-written).

We can trace characters during editing sessions, and design a view-only **auditing interface**.

The auditor can verify edits in the interface, performing a final check before the doc is sent/published.



InkSync Study 2: Study Conditions

Advice #4: Adapt the system for the study

Objective: Study whether Warn-Verify-Audit can assist participants in detecting and avoiding hallucinations.

Four conditions:

No Framework (Baseline)	Warn-Verify
Audit	Warn-Verify-Audit

Advice #4: Adapt the System for the Study

Problem: errors “only” occur in in 15–30% of documents, so not every participant would see a hallucination.

Solution: modify the system to produce **more hallucinations** during the study. Now each participant will encounter factual errors.

A usability study is not a product review.

Control the environment as much as you can.

☠ Advice #5: Prepare to Deal with Cheaters

Unavoidable: some % participants don't do the task well, either because: (1) they don't care, (2) they don't understand.

✔ Make it easy to spot unengaged participants.

Don't: "Force" participants to complete task (they'll find another way to cheat)

✔ Expect to filter out 10-20% of unengaged participants. Filter all data, don't cherry-pick.

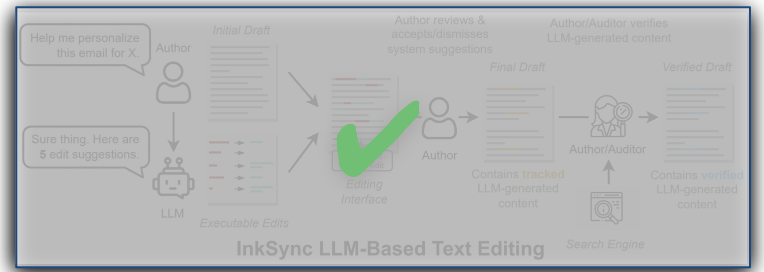
Don't: withhold payment of unengaged participant.
It's a part of the game

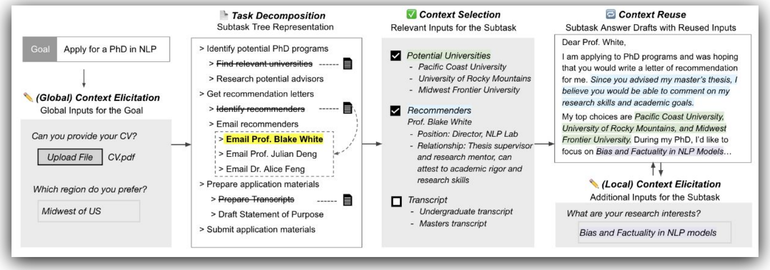
InkSync Study 2: Filter out any participant that doesn't run any verification during the study -> indicative of unengaged participant.

Overview

Part 1: An introduction to Human-Centered NLP

Part 2: Case studies with two human-centered systems





[InkSync](#) (Laban et al., 2024)

[JumpStarter](#) (Zhang and Wang et al., 2025)

JumpStarter: Human-AI Planning with Task-Structured Context Curation

[1]Xuanming Zhang*, [1]Sitong Wang*, [1]Jenny Ma, [2]Alyssa Hwang, [1]Zhou Yu, [1]Lydia Chilton

[1]Columbia University, [2]University of Pennsylvania

* denotes equal contribution

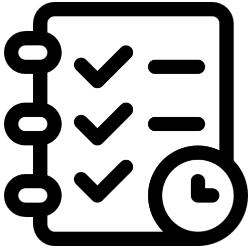
Everyone aspires to achieve personal goals

Moving to a new apartment

Starting a YouTube channel

Applying for PhD programs

However, getting started is often complex and daunting, especially for large projects



Planning

Action Initiation

Planning is challenging, because it requires breaking down tasks based not only on one's goals but also on their personal background

Goal: Apply for PhD in the US

Personal background: Are you an international student?

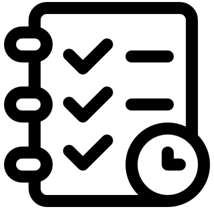
No

- Research programs
- Prepare for standardized tests
  - Take GRE if required
- Request recommendation letters
- Gather and prepare application materials
- Prepare for interviews

Yes

- Research programs
- Prepare for standardized tests
  - **Take TOEFL**
  - Take GRE if required
- Request recommendation letters
- Gather and prepare application materials
- **Visa application**

Action initiation is challenging, because while plans can be abstract, initiating an action requires deeply personal and relevant details



Intention to write
a request email
for rec letters

Be ready to write
a request email
for rec letters

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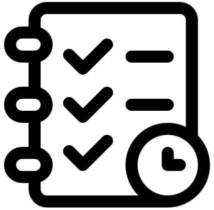


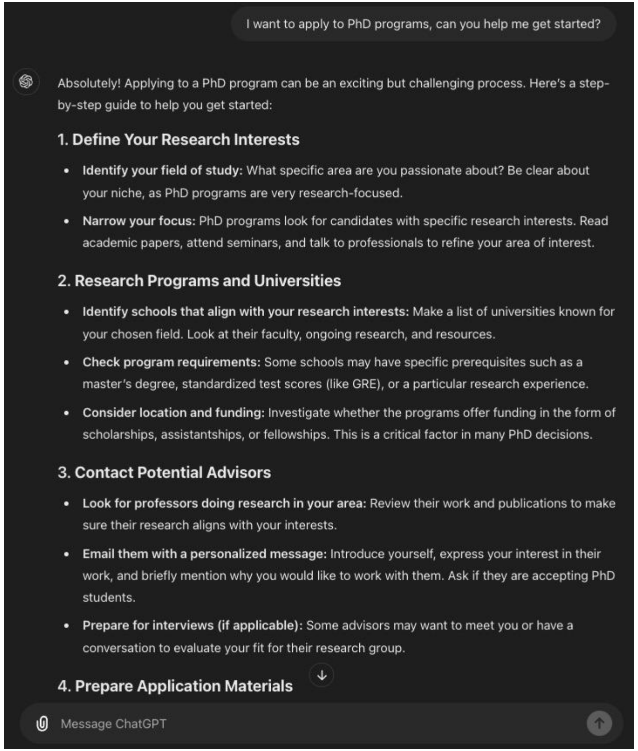
Intention to write
a request email
for rec letters

\+ *Current CV*
\+ *Programs to apply*
\+ *Skill highlights*
\+ *...*

Be ready to write
a request email
for rec letters

No worries, I can ask ChatGPT …

LLMs like ChatGPT have the potential to help, but often lacks the **personal context** and **structure** to be useful



- Personalization Deficiencies
  - Struggles to effectively elicit and integrate specific user backgrounds and needs
- Context Memory Limitations
  - Unable to retain past interactions, hindering consistency in project management
- Interface Constraints
  - Linear interface falls short in supporting systematic planning and progress tracking

Personalization Deficiencies

- Everyone has **unique** situations and requirements that need to be considered when embarking on personal projects.
    - E.g. “Applying for a PhD” -> consider *preferred research areas* and *personal background*.

- A complex goal like this must be further **decomposed** to become actionable.
    - E.g. “Requesting recommendation letters” needs further breakdown — one must first identify recommenders, gather supporting documents, send them **individualized** email drafts, and follow up accordingly

Context Memory Limitations

- ChatGPT allow users to provide context throughout their interactions
  - However, it can be **mentally demanding** for users to recall and articulate the necessary context relevant to the task and to frame the prompt accordingly.

- Ideally, ChatGPT would remember all the context provided by users in the chat and select the most appropriate ones to support each task.
  - However, this capability is limited by the inherent restrictions on **context window** size [1, 2, 3], and the process is **not transparent** to users

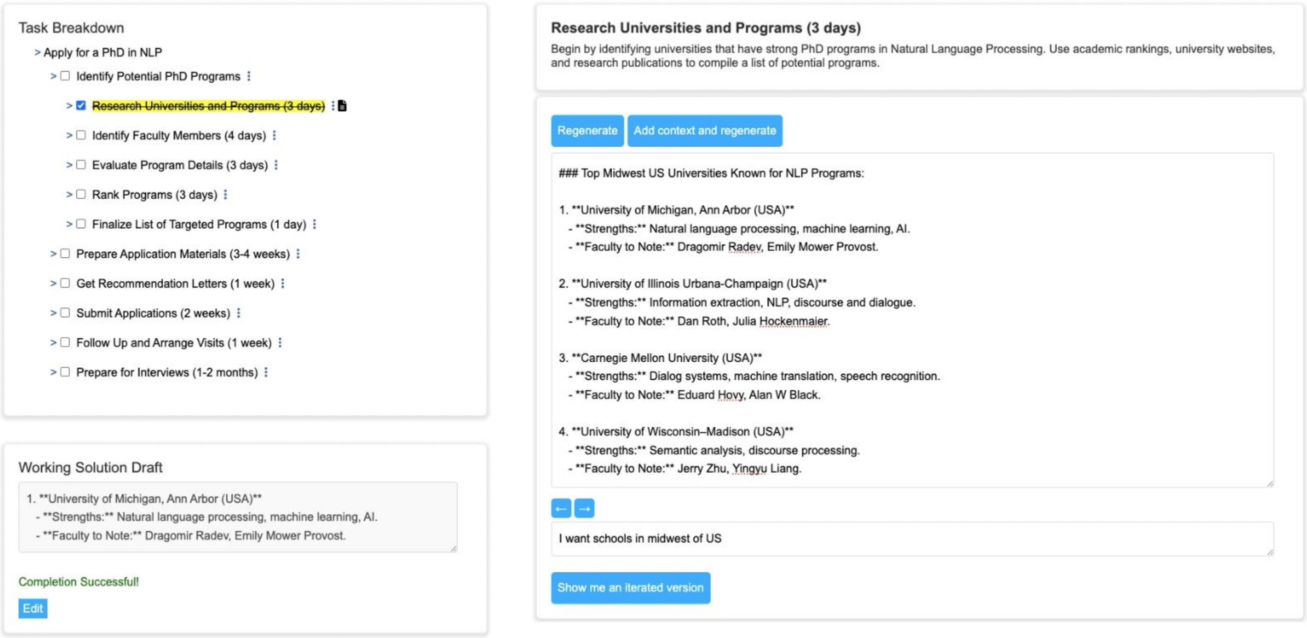
[1] Shouyuan Chen et.al, “Extending context window of large language models via positional interpolation.”, 2023
[2] Bowen Peng et.al, “Yarn: Efficient context window extension of large language models.”, 2023
[3] Howard Yen et.al, “Long-context language modeling with parallel context encoding”, 2024

Interface Constraints

- Linear chat interfaces like ChatGPT lack the **structure** to help people systematically plan their roadmap toward their goals.

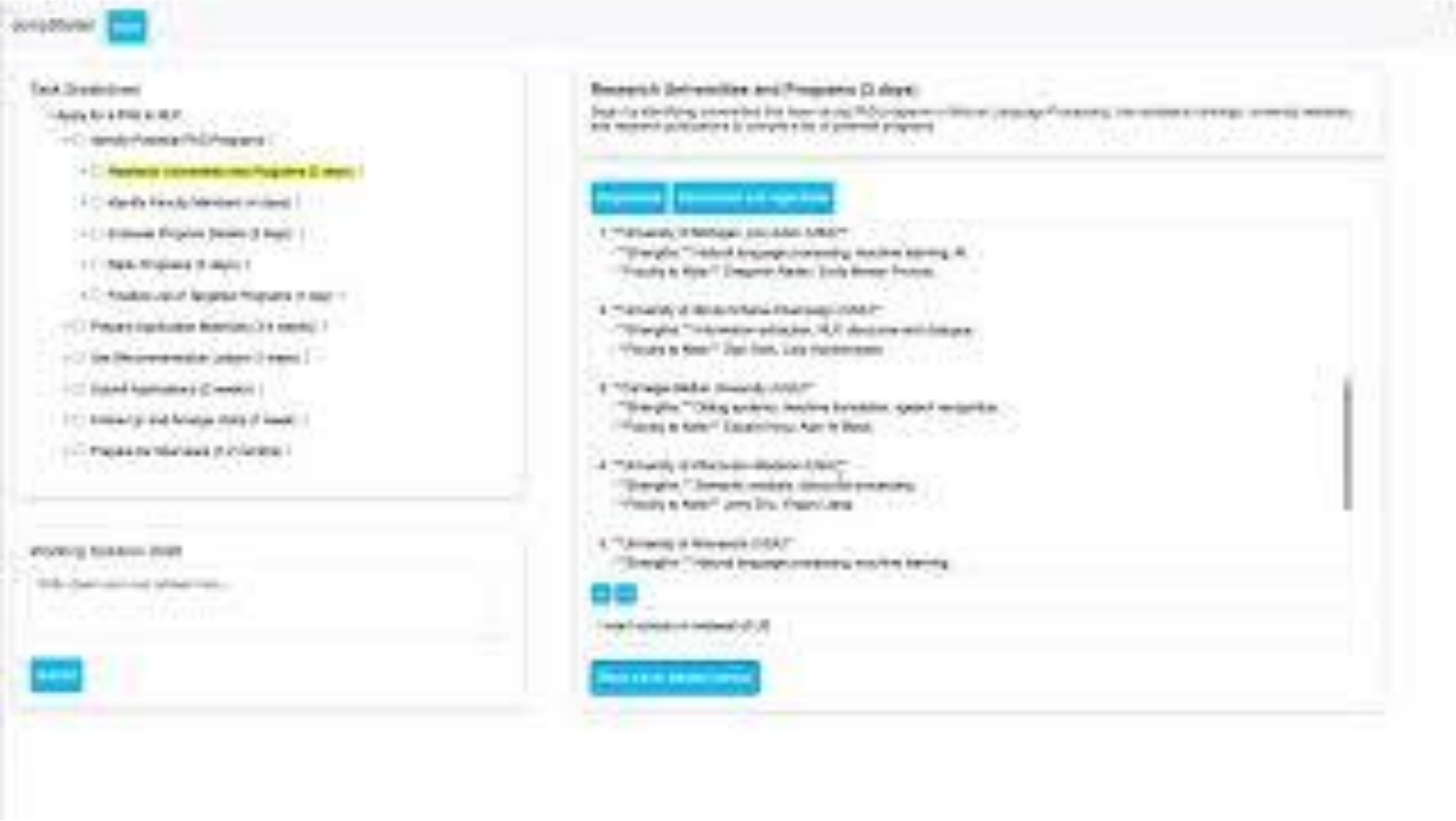
- Although people can ask ChatGPT to break down the steps further, it is easy to lose track of their progress and the context of each task
    - More details in our user studies!

To better support people in long-context planning, we introduce a system called JumpStarter



Through LLM-powered context curation, JumpStarter assists users by 1) decomposing tasks into hierarchical steps and 2) drafting working solutions.

System Demo



How does JumpStarter fit in the Design Space?

Aspect	Dimensions	Tags

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Task	Personal Project Planning and Action Initiation	Idea Generation, Planning, Drafting and Revision

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Task	Personal Project Planning and Action Initiation	Idea Generation, Planning, Drafting and Revision
User	Demographic Profile	Focused on goal-oriented people (have a goal in mind but don’t know how to get started)
User	Relationship to System	Agency (easy to accept/dismiss individual suggestions), Trust (Factuality Checking)

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Interaction	UI — Interface Paradigm	Tree structure: subtask suggestion / detection Text Editor: working solution drafting Pop-up window: context selection, elicitation
Interaction	UI — Visual Differentiation	Formatting (task highlighting, draft creation)
Interaction	UI — Initiation	Both Reactive (drafting) and Proactive (Subtask detection, Context Section / Elicitation)

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Interaction	UI — Visual Differentiation	Formatting (task highlighting, draft creation)
Interaction	UI — Initiation	Both Reactive (drafting) and Proactive (Subtask detection, Context Section / Elicitation)
Ecosystem	Model Access	Does not allow for model change
Ecosystem	Usability Consistency	Visual Resemblance to other Project Planning and Text Editing systems

Technical Evaluation - Subtask Detection

- Test suite
  - 20 test cases from real and diverse use cases, labeled by expert users.

Task and Description	Correct Answer
My user is working on the task Research Potential PhD Programs: User needs to start by researching various PhD programs in HCI. Identify the top programs of interest, considering factors like curriculum, faculty, past dissertations, program requirements, funding options, and university reputation. My user needs to know if the current task needs to be decomposed	Yes, it should be decomposed and it is not actionable
My user is working on the task Research on prerequisites for obtaining a US license: Familiarize yourself with the local laws and requirements for applying for a driver's license in New York, paying special attention to the requirements for foreign individuals with a non-US driver's license. My user needs to know if the current task needs to be decomposed.	No, it is actionable and it does not require task decomposition

Technical Evaluation - Subtask Detection

- Baselines
  - Zero-shot prompting
  - **Few-shot prompting**

Prompting techniques	Accuracy
Zero-shot	0.35
Few-shot	**0.58**

System prompt

You are a useful assistance to detect if the current task needs to be further decomposed if it is not actionable and the primary goal of the task can not be viewed as a singular, distinctive deliverable. Based on the user prompt, please output Yes if it needs to be decomposed; No otherwise meaning it is actionable and does not require task decomposition
Here are some examples:
User: My user is working on the task Research on Prospective Companies and Positions: Conduct a comprehensive search on potential companies and specific research scientist internship positions in the NLP field. Understand what each role entails, identify skill requirements, and evaluate how they align with your research interests. My user needs to know if the current task is specific and actionable
Answer: Yes

User: My user is working on the task Identify Potential Universities: Create a list of universities that offer PhD programs in HCI. The selection can be based on factors such as reputation, HCI research focus, published HCI research papers, faculty expertise etc. My user needs to know if the current task is specific and actionable.
Answer: No

User: My user is working on the task Identify Required Documents: Reseach and confirm all the necessary documents required for the non-driver ID application, ensuring to list all forms of acceptable proofs such as a birth certificate or passport for identity, Social Security Card or W-2 form for Social Security number, and utility bills or lease agreement for proof of residency. My user needs to know if the current task is specific and actionable.
Answer: No

Now, let's start prediction:

User prompt

My user is working on the task {task title}: {task description}. My user needs to know if the current task needs to be decomposed.

Technical Evaluation - Subtask Detection

- Few-shot + other prompting methods results

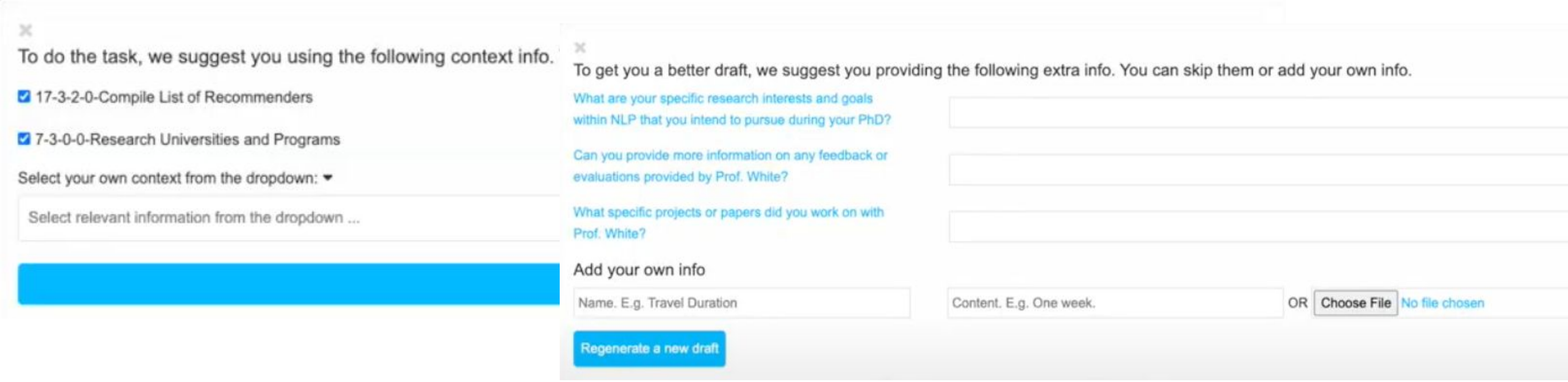
Prompting techniques	Accuracy		Statistics	
	Mean	SD	p	Sig.
Zero-shot	.35	.000		
Few-shot	.58	.040		
+ CoT	.62	.050	.405	-
+ CoT + Tree	.69	.020	.004	**
+ CoT + Draft	.72	.020	.009	**
+ CoT + Tree + Draft	.87	.040	.000	***

Table 1: The technical evaluation results for Subtask Detection comparing different prompting techniques, where the p-values (–: $p > .100$, +: $.050 < p < .100$, $*$: $p < .050$, $**$: $p < .010$, $***$: $p < .001$) are reported. Note that the p-values are computed against the few-shot-only baseline. Few-shot combined with CoT+Tree+Draft achieved the best accuracy.

- Generating the initial solution draft introduced a trade-off between **latency** and **accuracy**
- We opted for the slightly less performant **few-shot CoT with tree levels** for a better user experience.

Technical Evaluation - Context Curation

- We conducted a controlled within-subjects lab study for the following conditions
  - (1) Context dumping: context saving only -> what ChatGPT does
  - (2) Context filtering: context saving and context selection
  - (3) Context curation: context saving, context selection, and context



Technical Evaluation - Context Curation

Participant	Personal goal	Condition order
E1	Apply to a fellowship	(1) → (2) → (3)
E2		(2) → (3) → (1)
E3	Get a driver's license	(3) → (1) → (2)
E4		(1) → (2) → (3)
E5	Organize a team event	(2) → (3) → (1)
E6		(3) → (1) → (2)

Table 2: Overview of expert participants for the comparative study. Six experts were assigned one of three goals to evaluate under all three conditions, which were presented in shuffled order to avoid biasing the results.

We asked the participants to generate the **subtasks** and **working solutions** exactly once

They then rated the quality on a **seven-point Likert scale** while providing a verbal explanation for their choice.

Technical Evaluation - Context Curation

	Subtask quality	Working solution quality
(2) vs. (1)	$p = .436$ -	$p = .001$ **
(3) vs. (1)	$p < .001$ ***	$p < .001$ ***
(3) vs. (2)	$p < .001$ ***	$p < .001$ ***

Table 3: The statistical test results comparing three conditions, where the p-values (–: $p > .100$, +: $.050 < p < .100$, ∗: $p < .050$, ∗∗: $p < .010$, ∗ ∗ ∗: $p < .001$) are reported. Our full context curation method (3) outperformed context saving alone (1) and context saving and selection (2).

Technical Evaluation - Context Curation

- Baseline
  - Saving context only (what ChatGPT does)
- Context curation
  - Saving, selecting and eliciting context

	Baseline	**Context curation**	**p-value**
Plan quality	5.26	**6.12**	<.001 ***
Working solution quality	5.04	**6.36**	<.001 ***

E5: “Tended to produce only **general tips** for creating the itinerary”

Technical Evaluation - Context Curation

- Baseline
  - Saving context only (what ChatGPT does)
- Context curation
  - Saving, selecting and eliciting context

	Baseline	Context curation	p-value
Plan quality	5.26	**6.12**	<.001 ***
Working solution quality	5.04	**6.36**	<.001 ***

E1: “the email draft generated with context selection was more **personalized** than that created without it.”

Technical Evaluation - Context Curation

- Baseline
  - Saving context only (what ChatGPT does)
- Context curation
  - Saving, selecting and eliciting context

	Baseline	Context curation	p-value
Plan quality	5.26	**6.12**	<.001 ***
Working solution quality	5.04	**6.36**	<.001 ***

E4: "the elicitation questions posed at the beginning provided the right plan to start with."

Technical Evaluation - Context Curation

- E1:
  "*I like the recommendation letter request email draft it gives me, as it considers much of my* ***background*** *that I saved in the previous 'update your CV' task. I did the same thing in the previous round [context dumping] but did not feel it was as effective.*"

- E2:
  "*The subtasks were precise, fitting the* ***unique aspects of the fellowship I am applying to****, which requires only* ***one*** *recommendation letter, though typically more are needed.*"

User Study

- We conducted a user study with ten participants (average age=23.8, six female, four male), each exploring a personal goal with ChatGPT and JumpStarter

- Each participant were given a maximum of 25 minutes to complete each task with ChatGPT vs JumpStarter

User Study - Personal goals

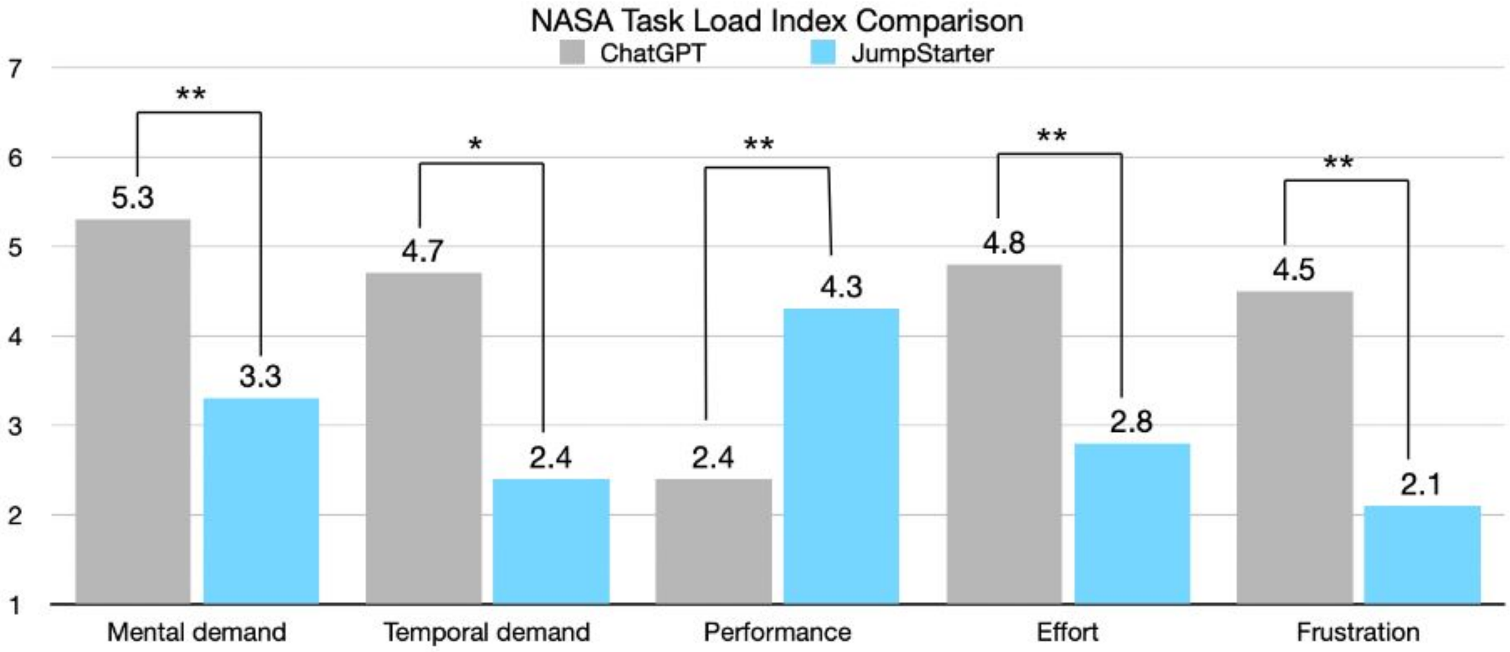
	Personal goal	Goal type
P1	Start a side job	Career
P2	Organize a weekly game night	Life
P3	Land a job offer	Career
P4	Prepare for the LSAT	Academia
P5	Manage social media accounts	Creativity
P6	Move to a new apartment	Life
P7	Create a portfolio website	Creativity
P8	Prepare to deliver a tutorial	Academia
P9	Start a new YouTube channel	Creativity
P10	Organize a family reunion	Life

User Study - Evaluation Metrics

- **Task load** - NASA Task Load Index dimensions [4]
    - *Mental demand*: How mentally challenging the task was.
    - *Temporal demand*: Time pressure experienced during the task.
    - *Performance*: How successful the person feels they were in accomplishing the task.
    - *Effort*: How hard the person had to work to accomplish the task.
    - *Frustration*: The amount of stress, irritation, or frustration experienced during the task

[4] Sandra G Hart. NASA-task load index (NASA-TLX), In Proceedings of the human factors and ergonomics society annual meeting, 2006

Compared to ChatGPT, JumpStarter significantly reduces users' task load



Compared to ChatGPT, JumpStarter significantly reduces users' task load

- P2:
  *"ChatGPT info dumps a lot, and I have to keep the structure in my brain, whereas JumpStarter gave me a **structure that I could easily follow**. "*

- P10:
  *"I appreciate the **questions** the system (JumpStarter) asked **when I felt stuck about how to iterate on drafting events** [for a family reunion]. Entering the ages of the family members actually provided many better choices."*

- P8:
  "In ChatGPT, the information load is high—I have to think very hard about what information I should provide in order to get things that work for me. "

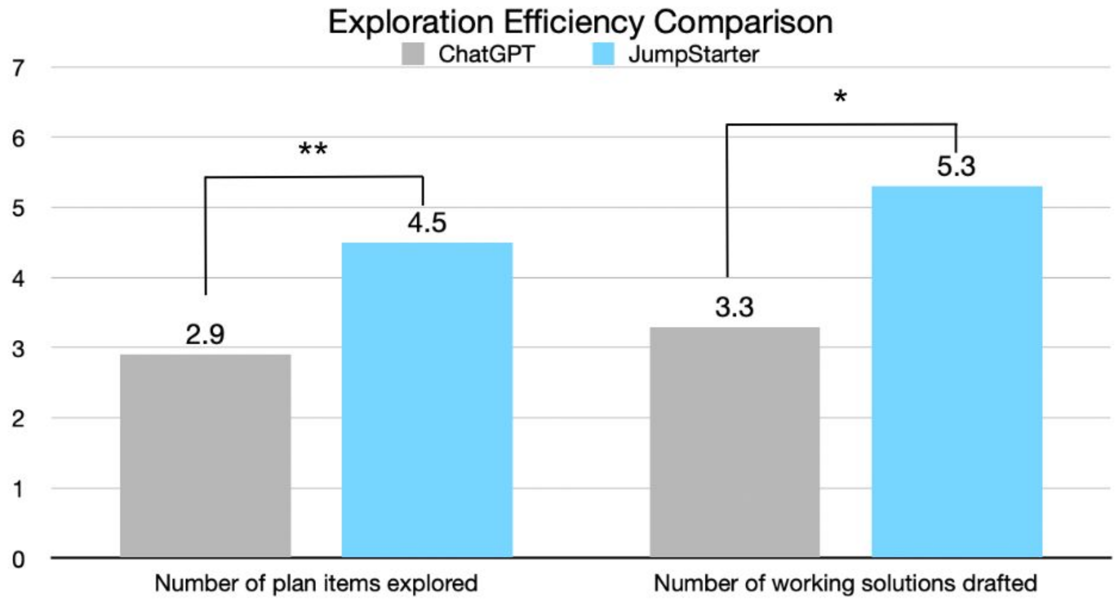
Compared to ChatGPT, JumpStarter significantly reduces users' task load

- P1 after asking ChatGPT to generate questions they could answer to improve the response
  "*hard to answer as there were many and they were a bit too abstract.*"

- P1 after answering all questions
  "*The response are still a bit too general and not useful.*"
  "*ChatGPT seemed to forget these eight answers soon after in my following chat with it* "

User Study - Evaluation Metrics

- **Exploration Efficiency**
  - Within 25 mins: the number of plan items explored + the number of working solutions drafted



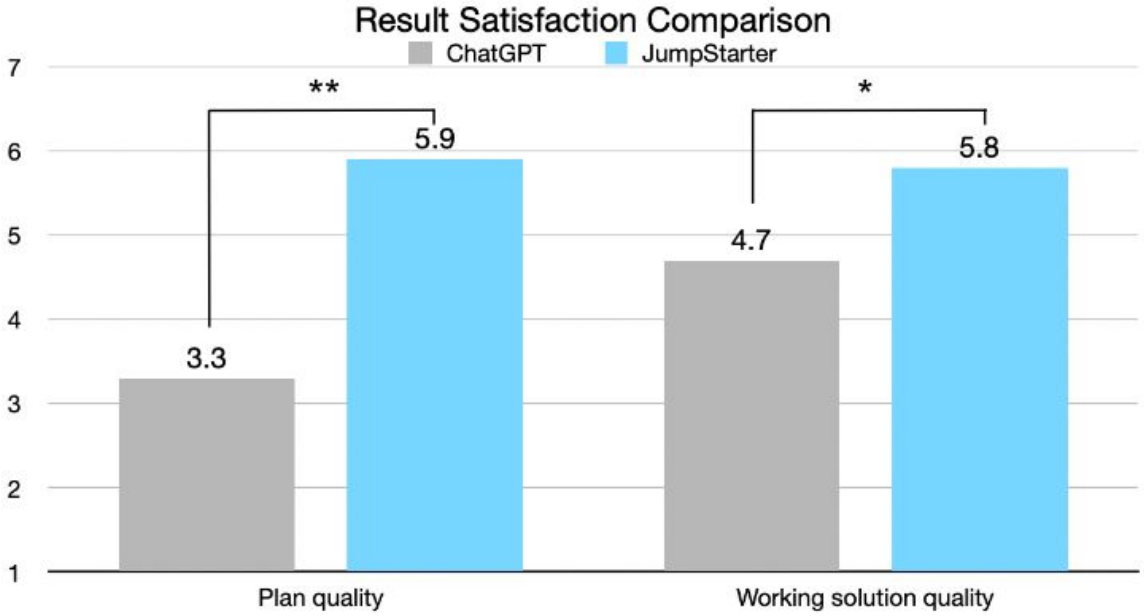
User Study - Exploration Efficiency

- In ChatGPT, it can be easy for users to become fixated on a single plan item, whereas JumpStarter helps users maintain an overview of the entire plan
  - P8: "*[With ChatGPT,] I went into a detailed level that was not necessary very quickly, and I just forgot to do the general planning.*"

- JumpStarter's structure enables users to jump between tasks and focus on one manageable task at a time, making it easier for them to arrive at a working solution
  - P4: " *feel JumpStarter has a more* ***flexible structure*** *compared to ChatGPT. I like that I can easily jump between tasks and pick the ones I care about the most.*"

User Study - Evaluation Metrics

- **Satisfaction level**
    - Plan quality + working solution quality

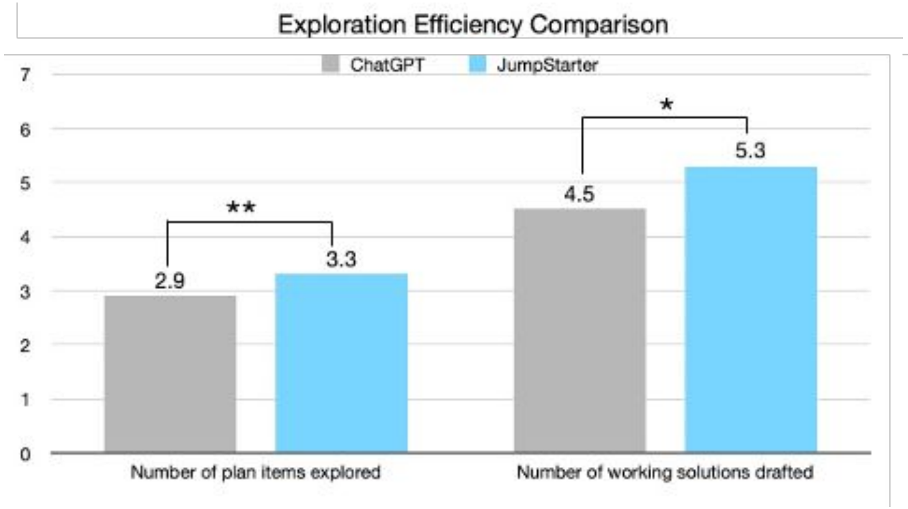


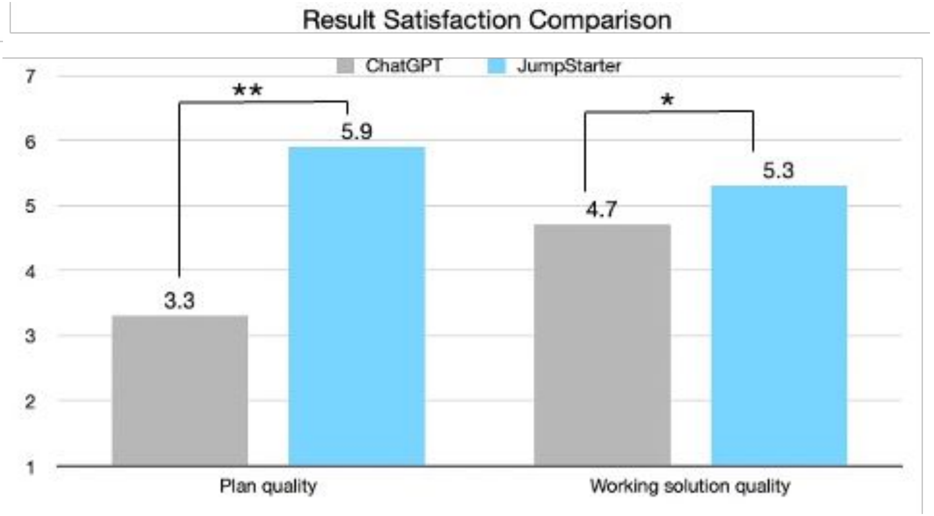
User Study - Satisfaction level

- JumpStarter often poses important questions to clarify users' needs and incorporates that information into its planning
  - P9 (who wants to start their YouTube channel):
    "*Identifying the target audience was not listed while I was using ChatGPT. I had to ask about it separately. Meanwhile, in JumpStarter, I was asked at the very beginning if I already knew my target audience. I said I needed help, and then it listed identifying the target audience as the first step in my plan, which was nice.* "

    "*JumpStarter asked about my familiarity with video editing at the very beginning, and I told them I was an expert. I feel that this was reflected well in the plan, like it assigned a shorter duration for the video editing task, which better suited my situation*"

Compared to ChatGPT, JumpStarter significantly increases users' exploration efficiency and satisfaction level





Blending Chat and Structure for Interacting with LLMs

- Our user study indicates that **a structured approach** is beneficial for handling more complex tasks

- Structure aids in generating **structured thoughts and personalized response** [6]

- A structured format helps users **more easily understand** the information provided by GPT, avoiding the overwhelming experience of endless scrolling [7]

[6] Xiao Ma et.al, Beyond chatbots: Explorellm for structured thoughts and personalized model responses. In CHI LBW 2024.
[7] Peiling Jiang et.al, Graphologue: Exploring large language model responses with interactive diagrams. In UIST 2023.

Blending Chat and Structure for Interacting with LLMs

- Does this mean that we should always prioritize structure over chat when using LLMs for complex problem-solving?
  - Users show an interest in engaging with a chatbot at certain stages of their exploration journey
    - E.g. when users encounter *unfamiliar terms* such as “research proposal” or “back-end coding”

- “Form fill-in” method for users to answer elicitation questions
  - A chat interaction feels more natural 🤔

LLM-powered Context Curation for Complex Problem Solving

- Our studies indicate that AI/LLMs is good at selecting and eliciting context.

- “Pull” mode in ChatGPT VS “Push” mode in JumpStarter
    - “Pull” mode: users had to identify all requirements and constraints themselves to formulate a prompt -> **Reactive**
    - “Push” mode: users can review questions posted by JumpStarter and select which ones to answer. -> **Proactive**

Recalling information is more cognitively demanding than **recognizing** it [5]

[5] Fergus IM Craik and Robert S Lockhart. 1972. Levels of processing: A framework for memory research. Journal of verbal learning and verbal behavior

LLM-powered Context Curation for Complex Problem Solving

- Our studies indicate that AI/LLMs is good at selecting and eliciting context.
- “Pull” mode in ChatGPT VS “Push” mode in JumpStarter
- Good sense of transparency and control
    - Users also appreciated the system assisting them in selecting context from the pool and transparently showing it before drafting a solution

LLM-powered (... em Solving

- Our studies inc ... iting context.
- “Pull” mode in
- Good sense of
  - Users also ... ing context from the pc ... a solution

Andrej Karpathy
@karpathy

+1 for "context engineering" over "prompt engineering".

People associate prompts with short task descriptions you'd give an LLM in your day-to-day use. When in every industrial-strength LLM app, context engineering is the delicate art and science of filling the context window with just the right information for the next step. Science because doing this right involves task descriptions and explanations, few shot examples, RAG, related (possibly multimodal) data, tools, state and history, compacting... Too little or of the wrong form and the LLM doesn't have the right context for optimal performance. Too much or too irrelevant and the LLM costs might go up and performance might come down. Doing this well is highly non-trivial. And art because of the guiding intuition around LLM psychology of people spirits.

On top of context engineering itself, an LLM app has to:
- break up problems just right into control flows
- pack the context windows just right
- dispatch calls to LLMs of the right kind and capability
- handle generation-verification UIUX flows
- a lot more - guardrails, security, evals, parallelism, prefetching, ...

So context engineering is just one small piece of an emerging thick layer of non-trivial software that coordinates individual LLM calls (and a lot more) into full LLM apps. The term "ChatGPT wrapper" is tired and really, really wrong.

Limitation and Future Work

- JumpStarter primarily assists users in figuring out “how” to achieve their personal goals

  -> the “why” problem—motivation (e.g. self-regulation or emotional challenges)

- JumpStarter was designed to support people with personal goals that primarily involve cognitive or knowledge work

  -> other personal goals: physical goals (i.e. losing weight), behavioral goals (e.g. overcoming shyness), and spiritual goals (e.g. coming to terms with one’s faith)

Limitation and Future Work

- JumpStarter utilizes GPT-4 as its core engine for providing information. It is prone to generate inaccurate or hallucinated information

  -> Integrate search agents to enhance credibility of the information provided

- JumpStarter currently requires users to manually add context to the system.

  -> Adopting a richer and more flexible method for capturing context that integrates into users' current workflows would be beneficial

- JumpStarter can only handle context in text form

  -> Expand to handle multimodal context!

Summary of today's lecture

- Human-centered NLP is beyond the NLP method

Situation: I have built a cool NLP model and I want to people to use it!
What should I consider when building the system that uses my NLP method?

⮡ Task, User, Interaction, Ecosystem, …

- Usability Testing

Situation: I have built a system that uses my NLP model.
How to test if "it works" for users? How do we know if users like it?

⮡ The core of Human-Centered systems: **User Studies**

Recap of Usability Study Advice

- #1: Consider the Task & Participants Carefully
- #2: Select Broad Study Conditions
- #3: Include Redundancy in Evaluation Signal
- #4: Adapt the System for the Study
- #5: Prepare to Deal with Cheaters

Attendance Question